Quantum Mechanics

Saavan Shah

June 2026

Contents

1 Introduction: **2**

2 The Stern Gerlach Experiment and Our First Look into Quantum Mechanics **3**

3 Postulates of Quantum Mechanics **4**

4 Spin Precession **6**

5 The Position and Momentum Bases **7**
5.1 The Position Basis . . . 8
5.2 The Momentum Basis . . . 8
5.3 Canonical Commutation Relation and the Momentum Operator in Position Space . . . 8
5.4 Momentum and Position Bases as Fourier Duals . . . 9

6 Potential Wells & Barriers **9**
6.1 The Free Particle . . . 9

7 The Infinite Square Well **10**
7.1 The Finite Square Well . . . 10
7.2 The Quantum Harmonic Oscillator . . . 11
7.2.1 Ladder Operators: . . . 11
7.2.2 The Energy Spectrum . . . 11
7.2.3 Eigenstates . . . 12
7.2.4 Matrix Representations . . . 12
7.3 The Potential Barrier and Tunneling . . . 13
7.4 Transmission and Reflection Coefficient . . . 13

8 Angular Momentum and the Hydrogen Atom **14**
8.1 Orbital Angular Momentum . . . 14
8.2 Ladder Operators and the Spectrum of $\hat{L}^2$ and $\hat{L}_z$ . . . 14
8.3 Spherical Harmonics . . . 15
8.4 The Hydrogen Atom . . . 15
8.4.1 The Radial Equation . . . 16
8.4.2 The Energy Spectrum and the Radial Wavefunctions . . . 16
8.5 The Spectrum and Atomic Transitions . . . 16

9 Commutation Relations & Time Evolution **17**
9.1 The Commutator . . . 17
9.2 Commutators and the Uncertainty Principle . . . 17
9.3 Compatible Observables & Observable Sets . . . 17
9.3.1 Complete Sets of Commuting Observables . . . 18
9.4 Constants of Motion . . . 18
9.5 The Heisenberg Picture . . . 18

10 Spin and Angular Momentum Addition **19**
10.1 The Total Angular Momentum . . . 19
10.2 The Allowed Values of Total Angular Momentum . . . 20
10.3 The Clebsh-Gordan Coefficients . . . 20
10.4 Addition of Two Spin-1/2 Particles . . . 20
10.5 Spin-Orbit Coupling . . . 21

11 Transformations and Symmetries **21**
11.1 Unitary Operators as Transformations & Their Generators . . . 21
11.2 Corresponding Symmetry Pairs . . . 22
11.3 Symmetries of the Hamiltonian and Conservation Laws . . . 22

12 Multi-Particle Systems **23**
12.1 Two-Particle Hilbert Spaces . . . 23
12.2 Separable and Entangled States . . . 23
12.3 The Exchange Operator . . . 24
12.4 Identical Fermions and the Slater Determinant . . . 24
12.5 Identical Bosons . . . 25
12.6 Spin and Spatial Degrees of Freedom for Two Electrons . . . 25
12.7 The Density Matrix and Reduced States . . . 25

13 EPR Paradox and Schrodinger's Cat **26**
13.1 The EPR Argument . . . 26
13.2 Bell's Theorem . . . 26
13.3 What Bell's Theorem Does and Doesn't Say . . . 27
13.4 Schrodinger's Cat . . . 27
13.5 The Problem with Measurement . . . 27

14 Perturbation Theory and Adiabatic Processes **28**
14.1 Time-Independent Non-Degenerate Perturbation Theory . . . 28
14.1.1 First-Order Corrections (Non-Degenerate, Time-Independent PT) . . . 29
14.1.2 Second-Order Corrections (Non-Degenerate, Time-Independent PT) . . . 29
14.2 Degenerate Time-Independent Perturbation Theory . . . 29
14.3 Time-Dependent Perturbation Theory . . . 30
14.4 Fermi's (Dirac's) Golden Rule . . . 30
14.5 The Adiabatic Theorem . . . 31
14.6 Berry's Phase . . . 31

15 A Few Words on Relativistic Quantum Mechanics **32**

16 Conclusion **33**

1 Introduction:

With the prerequisite mathematics settled in the previous notes, we can now start getting into the weeds of quantum physics. I am writing this entire document as a complete record of all of the fundamental concepts of quantum mechanics, so it may be in development as you read it. Quantum Mechanics is a large departure from the classical world. Where we used to have certainty in determinism, the quantum world is dictated by uncertainty, not in a poetic sense, but literally. It is quite commonly said that it is impossible to find comfort in the physics of quantum mechanics, so we will try to find comfort in the math.

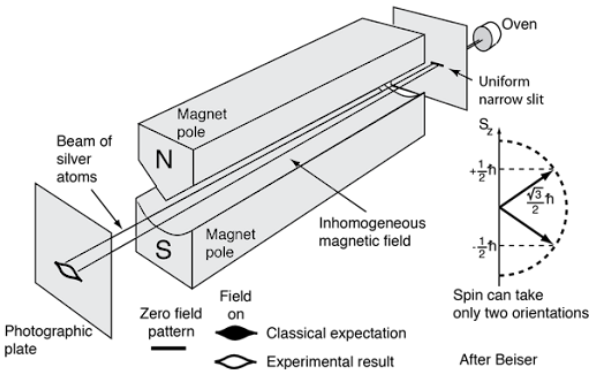


Figure 1: Stern Gerlach Experiment Setup

2 The Stern Gerlach Experiment and Our First Look into Quantum Mechanics

- The Stern-Gerlach experiment is a landmark 1922 physics experiment that proved the first experimental proof of quantization and the intrinsic angular moment of particles (dubbed electron spin)
- The experiment involves shooting a beam of neutral silver atoms through a strongly inhomogeneous (non-uniform) magnetic field and observing where they land on a detector screen.
- Classically, the charged particles/atoms behave like miniature bar magnets due to their magnetic moment: $\vec{\mu} = \frac{1}{2}q(\vec{v} \times \vec{r})$. However, since we are working with neutral silver atoms, the overall charge is zero. So, classically, the particles shouldn't have deflected, just continued dead on to hit the target.
- We observed that, rather, precisely half of the particles deflected upwards and half deflected downward, which suggests that we have an unknown quality of these silver atoms: spin, and it works probabilistically.
- We determined experimentally, the magnetic moment due to quantized spin is: $\vec{\mu} = \frac{1}{2m}qg\vec{S}$. And the force due to the magnetic field is $\vec{F} = -\vec{\nabla}U = \frac{-e}{2m}gS_z\partial_z B_z$.
- Through experiment, we determined that there are two possible values of S_z, $\pm\hbar/2/$ (or $\pm 1/2$ in natural units). We indicate the eigenstate associated with these measured values: $\pm\hbar/2 \rightarrow |\pm\rangle_z = |\pm\rangle$
- These eigenstates are with respect to the 1/2-Spin-z observable S_z. We can also measure eigenstates with respect to the 1/2-Spin-x or 1/2-Spin-y observable.
- Due to deep symmetry principles and representation theory, the spin matrices and eigenstates of a spin-1/2 particle are given by the Pauli Matrices (with a constant factor of $\hbar/2$ from earlier) and their eigenstates:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},\ \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix},\ \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$x - basis:\ |+\rangle_x = \frac{1}{\sqrt{2}}(|+\rangle_z + |-\rangle_z), |+\rangle_x = \frac{1}{\sqrt{2}}(|+\rangle_z - |-\rangle_z)$$

$$y - basis:\ |+\rangle_y = \frac{1}{\sqrt{2}}(|+\rangle_z + i\,|-\rangle_z),\ |-\rangle_y = \frac{1}{\sqrt{2}}(|+\rangle_z - i\,|-\rangle_z)$$

- Something you will notice is that an eigenstate in one basis for spin states will be a perfectly (50/50) random superposition in another spin basis. So, by leveraging measurement bases, we can put certain states "into" superpositions and "take others" out of superposition without measuring (I put quotes since we aren't collapsing any part of the wavefunction, i.e. measuring).

- We represent the spin state of the Stern-Gerlach experiment as a superposition; e.g. $|\psi\rangle = \frac{1}{\sqrt{2}}|+\rangle_z + \frac{1}{\sqrt{2}}|-\rangle_z$. As mentioned in the previous notes, the probability of measuring an eigenstate is given by the coefficient's modulus squared (Born Rule).
- Something to notice about the Born rule is that for our two-level system (a two-eigenstate system) to be in a 50/50 superposition, all we need is the modulus of the coefficient to be $1/\sqrt{2}$. Thus, $|\psi\rangle = \frac{1}{\sqrt{2}}(e^{i\theta}|+\rangle + e^{i\phi}|-\rangle$ is a valid superposition to describe our system, without more information. We can rewrite that as, $|\psi\rangle = e^{i\theta}(\frac{1}{\sqrt{2}}(|+\rangle + e^{i(\phi-\theta)}|-\rangle) = e^{i\theta}(\frac{1}{\sqrt{2}}(|+\rangle + e^{i\phi'}|-\rangle)$
- For a general superposition, we can write: $|\psi\rangle = e^{i\theta}(a|+\rangle + e^{i(\phi-\theta)}b|-\rangle) = e^{i\theta}(a|+\rangle + e^{i\phi'}b|-\rangle)$, where $a^2 + b^2 = 1$. We call the θ exponential the global phase, which we largely ignore, and we call the ϕ' exponential the relative phase, which we very much shouldn't ignore.
- If you look at the eigenstates in differing bases, you see they each differ by a relative phase, so dropping it by accident may switch your state altogether.

3 Postulates of Quantum Mechanics

- Quantum Mechanics is dictated by a series of postulates. These are necessary assumptions that we make in quantum mechanics to simulate how measured reality relates to the mathematics of quantum mechanics.
- In the previous sets of notes, we should have laid out the necessary framework to understand these axioms completely (up to certain definitions)
- **The Postulates of Quantum Mechanics**:
  1. Description of the State of a System (State and Observable Postulate)
  2. Measurement on a System
  3. Time Evolution of a System
  4. Composition Postulate
  5. Uncertainty Principle
  6. Symmetrization Postulate and the Pauli Exclusion Principle
- Let us now break down and define each of these postulates. We may need to review these later if it doesn't make sense when currently being stated.
- **Postulate I: Description of the State of a System**
  - The state of an isolated physical system is represented, at a fixed time t, by a normal (having norm 1) state vector $|\psi\rangle$ belonging to a Hilbert space $\mathcal{H}$ called the state space.
  - This state is unique up to a global phase.
  - Every measurable quantity is described by a Hermitian operator acting as an endomorphism on the state space. This operator is an observable, meaning that its eigenvectors form a basis for $\mathcal{H}$.
  - We can express our state vector as the superposition of eigenstates of relevant observables.
- **Postulate II: Measurement of a System:**
  - When we measure a physical quantity, the value must be one of the eigenvalues of the corresponding observable.
  - **Born Rule:** When a physical quantity is measured on a system in a normalized state, the probability of obtaining an eigenvalue of the corresponding observable is given by the amplitude squared of the appropriate wave function:
    1. For a unique eigenvalue: $\mathbb{P}(a_n) = |\langle a_n|\psi\rangle|^2$
    2. For a degenerate eigenvalue: $\mathbb{P}(a_n) = \sum^{g_n} |\langle a_n^i|\psi\rangle|^2$

3. For a continuous spectrum: $\mathbb{P}(a,b) = \int_a^b |\langle\alpha|\psi\rangle|^2 d\alpha$

- We can determine the expected value of a given observable (A) based on a state ket ($|\psi\rangle$) given by its weighted average: $\langle A\rangle_\psi = \sum_a (a(\sum^{g_n} |\langle a_n^i|\psi\rangle|^2))$ (for a discrete spectrum); $\langle A\rangle_\psi = \int_{-\infty}^{\infty} \alpha |\langle\alpha|\psi\rangle|^2 d\alpha$
- **Projection Operator**: is a linear operator $P : V \to V$ that satisfies $P^2 = P$. It maps each vector to its projection onto a subspace $W = \text{Im}(P)$, leaving vectors in W unchanged. Equivalently, every vector $v \in V$ can be decomposed as $v = P(v) + (v - P(v))$, where $P(v) \in \text{Im}(P)$ and $v - P(v) \in \ker(P)$.
- If we measure a physical quantity a_n from the state $|\psi\rangle$, then the state of the system immediately after the measurement is the normalized projection of $|\psi\rangle$ onto the eigen-subspace associated with a_n: $|\psi\rangle \implies P_n|\psi\rangle / \sqrt{\langle\psi|P_n|\psi\rangle}$

- **Postulate III: Time Evolution of a System**
  - The time evolution of the state vector $|\psi(t)\rangle$ is governed by the Schrodinger equation, where $H(t)$ is the observable associated with the total energy of the system (the Hamiltonian): $i\hbar\frac{d}{dt}|\psi(t)\rangle = H(t)|\psi(t)\rangle$.
  - We call the eigenvalue equation, $\hat{H}|\psi(t)\rangle = E|\psi(t)\rangle$, the time-independent Schrodinger Equation.
  - For a time-independent Hamiltonian, this becomes a solvable ODE, since the eigenstates of the Hamiltonian are time independent. As such, only the coefficients evolve with respect to time. In the energy basis, we have $|\psi(t)\rangle = \sum^n c_i(0)e^{i(E_i t/\hbar)}|E_i\rangle$
  - Something to notice: the probability of measuring a given energy value is constant with respect to time, but the relative phase changes, so the probability of measuring another observable may evolve with respect to time.
  - The time evolution of a closed system is described by a unitary transformation on the initial state: $|\psi(t)\rangle = U(t;t_0)|\psi(t_0)\rangle$

- **Postulate IV: the Composition Postulate**
  - The Hilbert space of a composite system is the Hilbert space tensor product of the state spaces associated with the component systems. For a non-relativistic system consisting of a finite number of distinguishable particles, the component systems are the individual particles.
  - For example, to consider the Hilbert space of the spin states of two independent distinguishable spin-1/2 particles, we can simply tensor the superposition of each particle to find the total state. Note: We must keep consistency in the order of tensors.
  - Suppose we have $|\psi_1\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$, $|\psi_2\rangle = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)$. We can represent this state as $|\psi_1\psi_2\rangle = \frac{1}{2}(|{+}{+}\rangle + |{-}{-}\rangle - |{+}{-}\rangle - |{-}{+}\rangle)$ within the total Hilbert space.
  - There are certain state vectors in the total Hilbert space that cannot be separated into two independent superpositions. E.g. $|\psi\rangle = \frac{1}{\sqrt{2}}(|11\rangle + |00\rangle)$, which is known as a Bell-state. This is known as an **entangled state**, and they are a major revelation of quantum mechanics.
  - Operators in this space are simply Kronecker products of operators, $A \otimes B$, where each operator only acts on their corresponding position (A acts on the first state and B acts on the second state).

- **Postulate V: The Uncertainty Principle**
  - For observables represented by Hermitian operators A and B, the uncertainties are quantified by their standard deviations ΔA and ΔB, where $\Delta C = \sqrt{\langle C^2\rangle_{|\psi\rangle} - \langle C\rangle^2_{|\psi\rangle}}$
  - The generalized uncertainty principle places a hard lower bound on how precise the error of two successive measurements can be. It states that

$$\Delta A\,\Delta B \geq \frac{1}{2}|\langle[A,B]\rangle|,$$

    where $[A,B] = AB - BA$ is the commutator of A and B.

- If two observables commute, i.e., $[A, B] = 0$, then the uncertainty relation places no fundamental lower bound on the product $\Delta A \, \Delta B$.
- The uncertainty relation arises from the Hilbert space structure of quantum mechanics and follows from the Cauchy–Schwarz inequality.
- The position-momentum uncertainty relation (arguably the most well-known),

$$\Delta x \, \Delta p \geq \frac{\hbar}{2},$$

is a special case obtained from the canonical commutation relation $[\hat{x}, \hat{p}] = i\hbar$.

- **Postulate VI: Symmetrization Postulate and the Pauli Exclusion Principle**
  - Identical particles are fundamentally indistinguishable and cannot be labeled by any physical measurement. Thus, we mandate that the state of a system of identical particles must be either symmetric or antisymmetric under the exchange of any two particles (depending on their spins).
  - For bosons, the wavefunction is symmetric under particle exchange: $\psi(\ldots, r_i, \ldots, r_j, \ldots) = \psi(\ldots, r_j, \ldots, r_i, \ldots)$. An example of a bosonic wavefunction is $\Psi_B(r_1, r_2) = \frac{1}{\sqrt{2}}[\phi_a(r_1)\phi_b(r_2) + \phi_b(r_1)\phi_a(r_2)]$, which is invariant under swapping r_1 and r_2. If we now have $\phi_a = \phi_b = \phi$, $\Psi_B(r_1, r_2) = \phi(r_1)\phi(r_2)$ (after renormalization)
  - For fermions, the wavefunction is antisymmetric under particle exchange: $\psi(\ldots, r_i, \ldots, r_j, \ldots) = -\psi(\ldots, r_j, \ldots, r_i, \ldots)$. An example of a fermionic wavefunction is $\Psi_F(r_1, r_2) = \frac{1}{\sqrt{2}}[\phi_a(r_1)\phi_b(r_2) - \phi_b(r_1)\phi_a(r_2)]$, which is anti-symmetric under swapping r_1 and r_2. If we now have $\phi_a = \phi_b = \phi$, $\Psi_B(r_1, r_2) = 0$.
  - This is precisely the **Pauli Exclusion Principle**. Any number of bosons may adopt the same wavefunction, but only one fermion can ever have a given wavefunction.
  - **Spin-Statistics Theorem**: Integer-spin particles are bosons, while half-integer-spin particles are fermions. Since all (non-anyonic) particles are either half-integer or integer spin, each particle must satisfy these relations.
  - The symmetrization postulate determines the allowed multiparticle states and underlies the statistical behavior of quantum systems. Bosons obey Bose–Einstein statistics and can occupy the same quantum state, whereas fermions obey Fermi–Dirac statistics and are subject to antisymmetry constraints.
  - The exclusion principle explains the shell structure of atoms and the organization of electrons into distinct energy levels. It is responsible for many macroscopic properties of matter, including atomic stability and the structure of white dwarf and neutron stars.
- Now that we have established all the necessary prerequisites and postulates, we are free to study all of Quantum Mechanics.

4 Spin Precession

- With the postulates established, we are ready to apply them to a concrete physical situation. Spin precession is perhaps the cleanest early application of Postulate III (time evolution), and it gives us our first dynamic picture of a quantum state, one that changes not only in phase, but in its observable properties over time.
- **A Spin-1/2 Particle in a Uniform Magnetic Field**:
  - Consider an electron placed in a uniform magnetic field oriented along the z-axis: $\vec{B} = B_0\hat{z}$. We established in the Stern-Gerlach section that the magnetic moment of such a particle is $\vec{\mu} = \frac{-e}{2m} g\vec{S} = \gamma\vec{S}$, where $\vec{S} = \langle S_x, S_y, S_z \rangle$ and γ represents the gyromagnetic ratio.
  - The energy of the system, as well known in Electromagnetism, is $H = -\gamma B_0 S_z = \hbar(\omega_0 \sigma_z)/2$. We call ω_0 the Larmor frequency (the natural frequency that our spin vector will precess)

- **Finding Energy Eigenstates**: Since H is proportional to σ_z, the energy eigenstates are precisely the eigenstates of σ_z, $|\pm\rangle$, and the energy eigenvalues are precisely $\pm\frac{\hbar}{2}\omega_0$. This makes physical sense: the state aligned with the field ($|+\rangle$) has higher energy, and the anti-aligned state ($|-\rangle$) has lower energy. Their separation is $\Delta E = \hbar\omega_0$.

- **Time Evolution of a General Spin State:**

  – Suppose at t=0, our particle is in a general normalized spin state: $|\psi(0)\rangle = cos(\frac{\theta}{2})\,|+\rangle + e^{i\phi}sin(\frac{\theta}{2})\,|-\rangle$. The angles θ and ϕ parameterize any point on the Bloch-Sphere (a visual way to understand two-level systems)

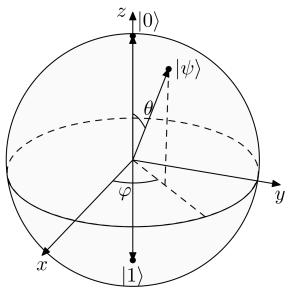


Figure 2: The Bloch Sphere

  – Since H is time-independent and we have expressed our initial state in the energy basis, we can use Postulate III.

$$|\psi(t)\rangle = cos(\frac{\theta}{2})e^{-i\omega_0 t/2}\,|+\rangle + e^{i\phi}sin(\frac{\theta}{2})e^{i\omega_0 t/2}\,|-\rangle \approx cos(\frac{\theta}{2})\,|+\rangle + e^{i(\omega_0 t+\phi)}sin(\frac{\theta}{2})\,|-\rangle$$

  – By dropping our overall global phase, we see that the state processes on the Bloch sphere in the direction of the applied magnetic field (θ is constant, but the azimuthal angle ϕ increases linearly in time at the rate of Larmor precession. This directly mirrors classical magnetic precession of classical magnetic moments.

  – If we were to write this in the x-basis, rather than the z-basis, we would get:

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}}((cos(\frac{\theta}{2}) + e^{i(w_0 t+\phi)}sin(\frac{\theta}{2}))\,|+_x\rangle + (cos(\frac{\theta}{2}) - e^{i(\omega_0 t+\phi)}sin(\frac{\theta}{2}))\,|-_x\rangle)$$

  – Here, we can clearly see that the probability of measuring an eigenstate changes in time. This is a significant revelation since if an observable doesn't share an eigenbasis with the Hamiltonian, the probability of measuring an eigenvalue will evolve with respect to time. If we were to now take the expected value of measuring S_x, we would also see it evolves with respect to time.

5 The Position and Momentum Bases

- With spin precession, we worked entirely in a finite-dimensional Hilbert space, a two-level system spanned by $|\pm\rangle$. Now, we will consider the continuous case with the position and momentum operators (an uncountably infinite basis set).

- This section builds up the position and momentum representations from scratch, makes sense of the wavefunction as a concrete object, and establishes the canonical commutation relation that is at the heart of the Heisenberg Uncertainty Principle.

5.1 The Position Basis

- We define the position operator $\hat{x}$ as the observable corresponding to the position of a particle. By Postulate I, its eigenstates must span the Hilbert space and correspond to definite measurement outcomes.
- Since position takes values across all of $\mathbb{R}$, we have a continuous family of eigenstates: $\hat{x}\,|x\rangle = x\,|x\rangle\,,\ x \in \mathbb{R}$. Here, $|x\rangle$ is the eigenstate in which the particle has a definite position x. These are our position basis kets.
- For this continuous basis, the orthogonality condition must be upgraded from the Kronecker delta, δ_{ij}, to the Dirac delta, $\delta(x - x') = \langle x|x'\rangle$, and the completeness condition becomes $\hat{I} = \int |x\rangle\langle x|\,dx$. Any state $|\psi\rangle$ can be expanded in this basis by inserting the identity: $|\psi\rangle = \hat{I}\,|x\rangle = \int |x\rangle\langle x|\psi\rangle\,dx = \int \psi(x)\,|x\rangle\,dx$
- The coefficient function $\psi(x) = \langle x|\psi\rangle$ is the wavefunction in the position representation. It is simply the projection of the state onto the position basis. Thus, the Born Rule becomes $\mathbb{P}(a,b) = \int |\psi(x)|^2 dx$. So, $|\psi(x)|^2$ is the probability density of finding the particle at position x (not the probability of measuring x).
- For a state to be normalizable, we require $\langle\psi|\psi\rangle = \int |\psi(x)|^2 dx = 1$. This means $\psi(x)$ must be square integrable.
- This contrasts with the concept of **Dirac normalization**, which applies to the basis kets $|x\rangle$ themselves. We established $\delta(x - x') = \langle x|x'\rangle$, not 1. This is the correct notion of "orthonormality" for continuous bases, since there is no way to normalize a position eigenstate to 1. This becomes important when we begin swapping bases.
- In the continuous bases, observables act directly on wavefunctions. In the position basis, the position operator simply multiplies the wave function by x: $\langle x|\hat{x}|\psi\rangle = x\,\langle x|\psi\rangle = x\psi(x)$
- The expectation value of position in a state $|\psi\rangle$ becomes the weighted integral $\langle\hat{x}\rangle = \langle\psi|\hat{x}|\psi\rangle = \int \psi^*(x)x\psi(x)dx$
- The momentum operator in the position basis is given by $\hat{p} = -i\hbar\partial_x$.

5.2 The Momentum Basis

- We define the momentum operator $\hat{p}$ as the observable for linear momentum. Just as with position, its eigenstates satisfy: $\hat{p}\,|p\rangle = p\,|p\rangle\,, p \in \mathbb{R}$, with orthogonality and completeness conditions defined similarly to the position bases. In fact, most of the principles are similar; we just swap a p for an x.
- The wavefunction in the momentum basis is: $\psi(p) = \langle p|\psi\rangle$, so $|\psi\rangle = \int \psi(p)\,|p\rangle\,dp$. We also define the probability density of measuring momentum p as $|\psi(p)|^2$.
- In the momentum basis, the momentum operator simply multiplies the wave function by p: $\langle p|\hat{p}|\psi\rangle = p\,\langle p|\psi\rangle = p\psi(p)$
- The position operator in the momentum basis is given by $\hat{x} = i\hbar\partial_p$.

5.3 Canonical Commutation Relation and the Momentum Operator in Position Space

- You may be wondering how we got the formula for the momentum operator in position space, so we will explore its roots and the underlying relationship between position and momentum as a canonical pair in this section.
- The relationship between position and momentum is not symmetric. In fact, this is one of the defining features of quantum mechanics. To find how $\hat{p}$ acts in the position representation, we use a key physical symmetry: translational invariance.

- Consider the translation operator T(a) which shifts a particle's position by a: $\hat{T}(a)\,|\psi\rangle = |\psi'\rangle$, where $\psi'(x) = \psi(x-a)$. For a very small translation, $0 < \epsilon <<< 1$, we can do a first order Taylor expansion: $\hat{T}(\epsilon)\psi(x) = \psi(x-\epsilon) \approx \psi(x) - \epsilon\frac{d\psi}{dx}$.

- So $\hat{T}(\epsilon) = \hat{I} - \epsilon\frac{d\psi}{dx}$. Since $\hat{T}(a)$ must be unitary (translations preserve norm), and following the same logic used to derive the Schrödinger equation from the time-evolution operator in the bridging notes, the generator of translations must be anti-Hermitian before multiplying by i.

- We identify: $\hat{T}(\epsilon) = \hat{I} - \frac{i}{\hbar}\epsilon\hat{p} \implies \hat{p} \leftrightarrow -i\hbar\frac{d}{dx}$. Which is precisely the momentum operator in the position basis.

- With this, we can compute the commutation relation on an arbitrary wavefunction $[\hat{x}, \hat{p}] = i\hbar$. This is the canonical commutation relation. It encodes the fundamental incompatibility of position and momentum as simultaneous observables and, via Postulate V, immediately implies the Heisenberg Uncertainty Principle.

5.4 Momentum and Position Bases as Fourier Duals

- We are now in a position to ask: what does a momentum eigenstate $|p\rangle$ look like in the position representation? We need to solve the eigenvalue equation $\hat{p}\,|p\rangle = p\,|p\rangle$ in position space or find $\langle x|p\rangle$: $\hat{p}\,\langle x|p\rangle = p\,\langle x|p\rangle \implies -i\hbar\frac{d}{dx}\langle x|p\rangle = p\,\langle x|p\rangle$. This is a simple first-order ODE, with solution (with a normalization factor) $\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}}e^{ipx/\hbar}$.

- This suggests that a momentum eigenstate in the position basis is a plane wave, a complex exponential extending over all space with wavelength $\lambda = h/p$. This is an elegant result: the position and momentum eigenstates are Fourier duals of one another. The inner product $\langle x|p\rangle$ is the Fourier kernel, which means the position and momentum wavefunctions are related by a Fourier transform.

$$\psi(p) = \langle p|\psi\rangle = \frac{1}{\sqrt{2\pi\hbar}}\int e^{-ipx/\hbar}\psi(x)dx$$

$$\psi(x) = \langle x|\psi\rangle = \frac{1}{\sqrt{2\pi\hbar}}\int e^{ipx/\hbar}\psi(p)dp$$

- The Heisenberg uncertainty principle is, in this sense, simply the statement that a function and its Fourier transform cannot both be sharply localized.

6 Potential Wells & Barriers

- Now, we have all the background we need to solve the time-independent Schrodinger Equation for a given potential: $\hat{H}(\psi(t)) = E(\psi(t))$.

$$[-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x)]\phi(x) = E\phi(x)$$

- Each version of this problem we will solve in this section will change the V(x) to a different function. In every case, our goal is the same: find the allowed energies E_n (the eigenvalue spectrum) and the energy eigenfunctions ($\varphi_n(x)$). All else, such as expected values and time evolution, we simply obtain from our other postulates.

6.1 The Free Particle

- The simplest potential we can do is $V(x) = 0$ everywhere. Thus, the TISE reduces to: $\frac{d^2\phi}{dx^2} = -(2mE/(\hbar^2))\phi$. This is an easily solvable ODE, with a general solution:

$$\phi(x) = Ae^{ikx} + Be^{-ikx}, k = \sqrt{2mE}/\hbar$$

- Unlike the bound-state problems below, there are no boundary conditions here to quantize k. Thus, every $k \in \mathbb{R}$ is allowed, so the energy spectrum is continuous: $E = \hbar^2k^2/(2m), k > 0$. This makes sense since a free particle can have any energy.

- Adding time dependence, the full solution becomes:

$$\phi(x) = Ae^{i(kx-\omega t)} + Be^{-i(kx+\omega t)},\ \omega = \hbar k^2/(2m)$$

- Observing this solution analytically, we see that the wavefunction is composed of a right-moving and left-moving plane wave. These are precisely the momentum eigenstates from Section 5.4, with $p = \hbar k$

- Something to notice: these plane wave solutions are not normalizable on all of $\mathbb{R}$, as discussed in Section 5; they are Dirac-normalized, not square-integrable. A physical free particle is represented by a wave packet: a superposition of plane waves weighted by some amplitude function $\varphi(k)$:

$$\psi(x,t) = \frac{1}{\sqrt{2\pi}} \int \phi(k) e^{i(kx-\omega t)} dk$$

- The wavepacket travels at the group velocity, $v_g = \frac{d\omega}{dk} = \hbar k/m$ (this matches classical p/m), and phase velocity, $v_p = \omega/k = \hbar k/(2m)$. Something to notice about phase velocity is that for a superposition of eigenstates, the phase velocity of each eigenstate is distinct, so over time, we will see the wavefunction "flatten."

7 The Infinite Square Well

- The infinite square well (particle in a box) is defined by: $V(x) = 0$ if $0 \leq x \leq L$ and $V(x) = \infty$ otherwise. Within the box, we have the same ODE as the free particle scenario. However, the difference is that we have implemented boundary conditions since outside the box, the wavefunction must be identically zero (requires infinite KE to maintain a non-zero amplitude).

- If we require that the wavefunction is continuous, the boundary conditions become $\phi(0) = 0$ and $\phi(L) = 0$. We will rewrite the solution from above using complex coefficients:

$$\phi(x) = Ae^{ikx} + Be^{-ikx} = A'sin(kx) + B'cos(kx)$$

- Applying $\phi(0) = 0$, forces B'=0. Applying $\phi(L) = 0$ then requires our eigenstates to fall into distinct modes: $sin(kL) = 0 \implies kL = n\pi, n \in \mathbb{N}$.

- This discretizes k, giving quantized energies: $E_n = (\hbar^2 k_n^2)/(2m) = (n^2\pi^2\hbar^2)/(2mL^2), n \in \mathbb{N}$. Consequently, we see that the eigenstates (normalized on $[0, L]$) are: $\varphi_n(x) = \sqrt{\frac{2}{L}} sin(\frac{n\pi x}{L})$

- The ground state energy $E_1 = (\pi^2\hbar^2)/(2mL^2)$ is nonzero. This is a consequence of uncertainty: confining a particle to a region of size L forces $\Delta x \sim L$, forcing $\Delta p \sim \hbar/L$ and a nonzero kinetic energy.

- The energies scale as n^2, so the spectrum gets increasingly spread out: $E_2 = 4E_1, E_3 = 9E_1, ...$ Also, The n-th eigenfunction will have exactly (n-1) nodes inside the well, and the opposite parity of n will determine the parity of the wavefunction (evenness or oddness).

- The eigenfunctions φ_n form a complete orthonormal, discrete orthonormal set on $[0, L]$: $\int_0^L \phi_m^*(x)\phi_n(x)dx = \delta_{mn}$. Any normalizable function on [0,L] can be expanded in this basis, which is actually a Fourier sine series in disguise (the eigenfunctions are Fourier modes).

7.1 The Finite Square Well

- As we progress with our understanding of potential wells, we arrive at our first "bizarre" case: The finite square well. You may be wondering what changes between the finite square well and the infinite square well. The short answer is that the wavefunction amplitude "leaks".

- The square well (particle in a box) is defined by: $V(x) = 0$ if $0 \leq |x| \leq a$ and $V(x) = V_0$ otherwise (I symmetrized our box for convenience). The particle can have total energy E in the range $0 < E < V_0$ (bound states) or $E > V_0$ (scattering states). We focus on bound states here.

- For a bound state with $0 < E < V_0$, define: $\kappa = \sqrt{2m(V_0 - E)}/\hbar > 0, l = \sqrt{2mE/\hbar} > 0$.
- We can break this up into two situations: Inside and Outside the Well
  - **Outside the Well** ($|x| > a$): the TISE gives $\ddot{\phi} = \kappa^2\phi$. Normalizable solutions must decay, so:
$$\phi(x) = \begin{cases} \pm Fe^{-\kappa x} & \text{if } x > a \\ Ge^{\kappa x} & \text{if } x < -a \end{cases} \tag{1}$$
  - **Inside the Well** ($|x| \leq a$): the TISE gives $\ddot{\phi} = -l^2\phi$, so $\phi(x) = Asin(lx) + Bcos(lx)$.
- Since $V(x)$ is an even function, we can choose eigenfunctions to be either even or odd (due to the parity symmetry of the Hamiltonian):
  - Even solutions: $\phi_{in}(x) = Bcos(lx)$ matched to ϕ with $F > 0,\ G > 0$
  - Odd solutions: $\phi_{in}(x) = Asin(lx)$ matched to ϕ with $F < 0,\ G > 0$.
- **Boundary Matching**: Matching φ and $d\varphi/dx$ continuously at x=a (continuous and smooth). For even solutions: $Bcos(la) = Fe^{-\kappa a}, -Blsin(la) = -F\kappa e^{-\kappa a} \implies ltan(la) = \kappa$. Doing a similar procedure for the odd solutions, we get $-lcot(la) = \kappa$. These are transcendental equations, but we can analyze them graphically. Let $z = la$ and $z_0 = a\sqrt{2mV_0}/\hbar$; the even condition then becomes: $tan(z) = \sqrt{(z_0/z)^2 - 1}$.
- The number of bound states depends on the depth and width of the well. There is always at least one even bound state, regardless of how shallow the well is.
- Some notes: The classical well would not have states "leak" out of the well with exponential solutions. Moreover, if we take the limit as $V_0 \to \infty$, we recover the infinite square well example.

7.2 The Quantum Harmonic Oscillator

- The harmonic oscillator potential is $V(x) = \frac{1}{2}m\omega^2x^2$, giving the Hamiltonian: $\hat{H} = \hat{p}^2/2m + m\omega^2\hat{x}^2/2$. This is a very important potential in physics because it is the leading order approximation of any smooth potential near a stable equilibrium point (by Taylor Expansion).

7.2.1 Ladder Operators:

- Rather than solving the Schrodinger ODE directly, we factor the Hamiltonian algebraically. The key idea is to write $\hat{H}$ in terms of two conjugate operators. We define:
$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\hat{x} + \frac{i}{\sqrt{2m\omega\hbar}}\hat{p},\ \hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\hat{x} - \frac{i}{\sqrt{2m\omega\hbar}}\hat{p}$$
- These are the lowering ($\hat{a}$) and raising operators($\hat{a}^\dagger$) (or ladder operators), with $\hat{a}^\dagger = (\hat{a})^*$. With these, we rewrite the Hamiltonian as $\hat{H} = \hbar\omega(\hat{a}^\dagger\hat{a} + 1/2) = \hbar\omega(\hat{N} + 1/2)$.
- The Hamiltonian is diagonal in the eigenbasis of $\hat{N}$ (the number operator). Note: the number operator is not $\hat{a}\hat{a}^\dagger$, since $[\hat{a}, \hat{a}^\dagger] = 1 \neq 0$. Moreover, the commutators with respect to $\hat{H}$ is: $[\hat{H}, \hat{a}] = -\hbar\omega\hat{a},\ [\hat{H}, \hat{a}^\dagger] = \hbar\omega\hat{a}^\dagger$

7.2.2 The Energy Spectrum

- Suppose we have $|n\rangle$ is an eigenstate of $\hat{N}$. We can determine that $\hat{a}\,|n\rangle$ and $\hat{a}^\dagger\,|n\rangle$ are also eigenstates shifted by ± 1: $\hat{H}(\hat{a}\,|n\rangle) = (\hat{a}\hat{H} - \hbar\omega\hat{a})\,|n\rangle = (E_n - \hbar\omega)\hat{a}\,|n\rangle$. So $\hat{a}\,|n\rangle$ is an eigenstate with energy $E_n - \hbar\omega$; $\hat{a}$ lowers the energy by one "rung," and $\hat{a}^\dagger$ raises it by one "rung."
- This gives us a ladder of equally spaced energy levels. But the ladder must terminate at the bottom: energy cannot be negative for the harmonic oscillator (since $\hat{H}$ is a sum of squares of Hermitian operators. So, there must exist a ground state $|0\rangle$ that $\hat{a}$ annihilates: $\hat{a}\,|0\rangle = 0$. Thus, it must have a ground state energy $E_0 = \hbar\omega/2$ and, in general, $E_n = \hbar\omega(n + 1/2), n \in \mathbb{N}$
- The energy levels are equally spaced with spacing $\hbar\omega$, which is a distinctive feature of the harmonic oscillator not shared by the square well.

7.2.3 Eigenstates

- The normalized action of the ladder operators on energy eigenstates is: $\hat{a}\,|n\rangle = \sqrt{n}\,|n-1\rangle\,, \hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle\,.$ The factors of $\sqrt{n}$ and $\sqrt{n+1}$ follow from requiring $\langle n|n\rangle = 1$ and using $\hat{a}^\dagger\,|n\rangle = n\,|n\rangle$. All higher eigenstates can be generated from the ground state by repeated application of $\hat{a}^\dagger$: $|n\rangle = (\hat{a}^\dagger)^n\,|0\rangle\,/\sqrt{n!}$

- The ground-state wavefunction $\varphi_0(x) = \langle x|0\rangle$ is found by solving $\hat{a}\,|0\rangle = 0$ in position space:

$$(\sqrt{\frac{m\omega}{2\hbar}}x + \frac{\hbar}{\sqrt{2m\omega\hbar}}\frac{d}{dx})\phi_0(x) = 0 \implies \frac{d\phi_0}{dx} = -\frac{m\omega}{\hbar}x\phi_0$$

- This separable ODE has solution: $\phi_0(x) = (\frac{m\omega}{\pi\hbar})^{1/4}e^{-m\omega x^2/2\hbar}$ The ground state is a **Gaussian**, the minimum-uncertainty wave packet, "saturating" $\Delta x\,\Delta p = \hbar/2$ (since Gaussians are invariant under Fourier transforms). Higher eigenstates are Gaussians multiplied by Hermite polynomials $H_n(x)$, which emerge when $\hat{a}^\dagger$ is applied repeatedly in position space.

- Something to notice: the equally spaced spectrum means we can think of each energy quantum $\hbar\omega$ as a discrete excitation.

7.2.4 Matrix Representations

- Since the energy eigenstates $\{|n\rangle\}$ form an orthonormal basis for the harmonic oscillator Hilbert space, we can represent every operator as a matrix in this basis using $O_{mn} = \langle m|\,\hat{O}\,|n\rangle$. The matrix elements of the ladder operators follow directly from $\hat{a}\,|n\rangle = \sqrt{n}\,|n-1\rangle$ and $\hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle$, giving

$$\hat{a} \longleftrightarrow \begin{pmatrix} 0 & \sqrt{1} & 0 & 0 & \cdots \\ 0 & 0 & \sqrt{2} & 0 & \cdots \\ 0 & 0 & 0 & \sqrt{3} & \cdots \\ 0 & 0 & 0 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \qquad \hat{a}^\dagger \longleftrightarrow \begin{pmatrix} 0 & 0 & 0 & 0 & \cdots \\ \sqrt{1} & 0 & 0 & 0 & \cdots \\ 0 & \sqrt{2} & 0 & 0 & \cdots \\ 0 & 0 & \sqrt{3} & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

  Note that $\hat{a}^\dagger = (\hat{a})^\dagger$ as matrices (i.e. the conjugate transpose), confirming that they are indeed Hermitian conjugates of one another. $\hat{a}$ is a subdiagonal matrix and $\hat{a}^\dagger$ is a superdiagonal matrix, each with entries $\sqrt{n}$ on the relevant off-diagonal.

- The number operator $\hat{N} = \hat{a}^\dagger\hat{a}$ is diagonal in the energy basis by construction, with eigenvalue n on the n-th diagonal entry:

$$\hat{N} \longleftrightarrow \begin{pmatrix} 0 & 0 & 0 & 0 & \cdots \\ 0 & 1 & 0 & 0 & \cdots \\ 0 & 0 & 2 & 0 & \cdots \\ 0 & 0 & 0 & 3 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

  One can verify this directly by computing $\hat{a}^\dagger\hat{a}$ as a matrix product: the sub- and super-diagonal factors combine via $(\sqrt{n})^2 = n$ to produce the diagonal entries. The Hamiltonian matrix then follows immediately from $\hat{H} = \hbar\omega(\hat{N} + \frac{1}{2})$:

$$\hat{H} \longleftrightarrow \hbar\omega \begin{pmatrix} \frac{1}{2} & 0 & 0 & 0 & \cdots \\ 0 & \frac{3}{2} & 0 & 0 & \cdots \\ 0 & 0 & \frac{5}{2} & 0 & \cdots \\ 0 & 0 & 0 & \frac{7}{2} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

  The diagonal entries are precisely $E_n = \hbar\omega(n + \frac{1}{2})$, and the off-diagonal entries are all zero, confirming that the $|n\rangle$ are energy eigenstates.

- We can also express the position and momentum operators in terms of our ladder operators:

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(a + a^\dagger),\ \hat{p} = -i\sqrt{\frac{m\hbar\omega}{2}}(a - a^\dagger)$$

7.3 The Potential Barrier and Tunneling

- We will now consider the exact opposite of a well, a wall. Consider a rectangular potential barrier: $V(x) = V_0$ if $0 \leq x \leq a$ and $V(x) = 0$ otherwise. A particle is incident from the left with energy E. Classically, if $E < V_0$, our particle is reflected; if $E < V_0$, it passes through. Quantum mechanics gives something richer in either case. We treat the two cases separately.

  - **Above Barrier Scattering** ($E > V_0$). Defining: $k = \frac{\sqrt{2mE}}{\hbar}$, $l = \frac{\sqrt{2m(E-V_0)}}{\hbar}$. We get the general solution:

$$\phi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx} & \text{if } x < 0 \\ Ce^{ilx} + De^{-ilx} & \text{if } 0 \leq x \leq a \\ Fe^{ikx} & \text{if } x > a \end{cases} \tag{2}$$

    We set the coefficient of e^{-ikx} in region III to zero: there is no left-moving wave to the right of the barrier (nothing to reflect off of from that side). Applying continuity and smoothness at $x = 0, a$ gives four equations in five unknowns (A,B,C,D,F). We normalize by setting A=1, treating it as the incident amplitude.

  - **Tunneling** ($E < V_0$): In this regime, l becomes imaginary. Define instead $\kappa = \sqrt{(2m(V_0 - E)}$ $/\hbar > 0$, so inside the barrier: $\phi(x) = Ce^{\kappa x} + De^{-\kappa x}, 0 \leq x \leq a$. The wavefunction does not oscillate inside the barrier — it decays (or grows) exponentially. The D term gives the dominant decaying solution, and C gives the growing one. Both must be kept since the barrier has a finite width. The structure outside the barrier is unchanged: incident + reflected on the left, transmitted only on the right. Continuity at x = 0 and x = a again gives four junction conditions. Note, the taller and wider the barrier, the lower the coefficient of F is, and the higher the coefficient for B.

7.4 Transmission and Reflection Coefficient

- To quantify how much of the wave gets through, we use the probability current. For a wavefunction $\psi(x,t)$, the probability current is: $J(x,t) = \frac{\hbar}{2mi}(\psi^*\partial_x\psi - \psi\partial_x\psi^*$. This satisfies the continuity equation $\partial|\psi|^2/\partial t + \partial J/\partial x = 0$, which is the conservation of probability in differential form.

- For a plane wave $\psi = Ae^{ikx}$, the current is $J = v|A|^2$, so the current carried by a wave is its speed times probability density.

- We define the **Reflection Coefficient** (R) and **Transmission Coefficient** (T) as the ratios of reflected and transmitted currents to their incident current:

$$R = \frac{|J_{ref}|}{|J_{inc}|} = \frac{|B|^2}{|A|^2},\ T = \frac{|J_{trans}|}{|J_{inc}|} = \frac{|F|^2}{|A|^2}$$

- Conservation of probability requires $R + T = 1$. which serves as a useful consistency check.

- Imagine an unbound state, ($E > V_0$), above a finite barrier. Solving the junction conditions and eliminating C and D gives:

$$T = \frac{1}{1 + \frac{V_0^2 sin^2(la)}{4E(E-V_0)}}, R = 1 - T$$

- Something to notice: even when the particle has more than enough energy to classically pass over the barrier, $T < 1$ in general. There is always some reflection from the edges of the barrier, a purely quantum effect with no classical analog.

- Now, imagine ($E < V_0$), a tunneling scenario, the same procedure with the decaying solutions gives:

$$T = \frac{1}{1 + \frac{V_0^2 sinh^2(\kappa a)}{4E(V_0 - E)}}, R = 1 - T$$

- The exponential factor $e^{-2\kappa a}$ is the dominant dependence: transmission decreases exponentially with both the barrier width a and the square root of the energy deficit $(V_0 - E)$. This is quantum tunneling; the particle has a nonzero probability of appearing on the far side of a classically forbidden region.

- Another thing to view is that there are relative maximum and minimum transition rates depending on the sine-squared term, which is an odd quantum effect.

8 Angular Momentum and the Hydrogen Atom

- The Heisenberg model of the hydrogen atom was one of the crowning jewels of (non-relative) quantum mechanics. It is one of the only physically realistic systems with an exact closed-form solution, and it introduces the machinery of angular momentum that underlies atomic structure and spin-orbit coupling. To solve the Hydrogen atom, we will use operator algebra (ladder operators).

8.1 Orbital Angular Momentum

- Classically, the angular momentum of a particle is $\vec{L} = \vec{r} \times \vec{p}$. Promoting these to quantum operators, we get $\hat{L} = \hat{r} \times \hat{p}$. In components, we get $\hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y, \quad \hat{L}_y = \hat{z}\hat{p}_x - \hat{x}\hat{p}_z, \quad \hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x$. The fundamental commutation relations of the angular momentum components follow from $[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}$ and a straightforward (if tedious) calculation. This yields the defining commutation relation behind angular momentum, $[\hat{L}_i, \hat{L}_j] = i\hbar\epsilon_{ijk}\hat{L}_k$, and defines all observables satisfying this relation as angular momentum. This yields the result that spin is a "type" of angular momentum.

- Because no two components of $\hat{L}$ commute, they do not share a simultaneous eigenbasis. However, the total angular momentum squared, $\hat{L}^2 = \hat{L}_x^2 + L_y^2 + L_z^2$, commutes with each component. So, $\hat{L}^2$ and $\hat{L}_z$ (by convention) share an eigenbasis: $\{|l, m\rangle\}$.

8.2 Ladder Operators and the Spectrum of $\hat{L}^2$ and $\hat{L}_z$

- We define the angular momentum ladder operators, in direct analogy to the harmonic oscillator: $\hat{L}_\pm = \hat{L}_x \pm i\hat{L}_y$. We obtain the following commutation relations: $[\hat{L}_z, \hat{L}_\pm] = \pm\hbar\hat{L}_\pm$, $[\hat{L}^2, \hat{L}_\pm] = 0$. The second relation tells us that ladder operators don't change total angular momentum but only the z-component of the angular momentum.

- Suppose $|l, m\rangle$ satisfies (by inference): $\hat{L}^2 |l, m\rangle = \hbar^2\lambda |l, m\rangle$ and $\hat{L}_z |l, m\rangle = \hbar m |l, m\rangle$. Then, $\hat{L}_\pm |l, m\rangle$ is an eigenstate of L_z with eigenvalue $\hbar(m \pm 1)$, while the eigenvalue of $\hat{L}^2$ is unchanged. So the ladder operators climb through a sequence of states with the same λ but increasing or decreasing m. (We will soon define λ)

- The ladder must eventually terminate. Since $\hat{L}^2 = \hat{L}_z^2 + \frac{1}{2}(\hat{L}_+\hat{L}_- + \hat{L}_-\hat{L}_+)$. Therefore, we need that $\lambda \geq m^2$. So for fixed λ there must be a max (m_{max}) and min (m_{min}) with: $\hat{L}_+ |l, m_{max}\rangle = 0, \hat{L}_- |l, m_{min}\rangle = 0$. This directly implies that $\lambda = l(l+1)$ and $m_{max} = -m_{min}$.

- $\hat{L}^2 |l, m\rangle = \hbar^2 l(l+1) |l, m\rangle$ and $\hat{L}_z |l, m\rangle = \hbar m |l, m\rangle$, where $l = 0, 1/2, 1, 3/2...$ and $m = -l, -l+1, ..., l-1, l$. For orbital angular moment, specifically, the single-valuedness of the wavefunction restricts $l \geq 0 \in \mathbb{Z}$. But for spin, we are allowed half-spins (as stated in section II).

- This entire process shows the power of derivations purely through commutation relations and operator algebra.

8.3 Spherical Harmonics

- To use angular momentum in the hydrogen atom, we need the wavefunctions $\langle \theta, \phi | l, m \rangle$ in position space. Since the hydrogen potential is spherically symmetric, it is natural to work in spherical coordinates (r, θ, ϕ), where:

$$x = r\sin\theta\cos\phi, \qquad y = r\sin\theta\sin\phi, \qquad z = r\cos\theta.$$

  In these coordinates, $\hat{L}_z$ and $\hat{L}^2$ take the form:

$$\hat{L}_z = -i\hbar\frac{\partial}{\partial\phi},$$

$$\hat{L}^2 = -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right].$$

- The simultaneous eigenfunctions of $\hat{L}^2$ and $\hat{L}_z$ are the **spherical harmonics** $Y_l^m(\theta, \phi)$, derived by Legendre Polynomials, defined by:

$$\hat{L}^2 Y_l^m(\theta,\phi) = \hbar^2 l(l+1)\, Y_l^m(\theta,\phi), \qquad \hat{L}_z Y_l^m(\theta,\phi) = \hbar m\, Y_l^m(\theta,\phi).$$

  They factor as $Y_l^m(\theta,\phi) = \Theta(\theta)\Phi(\phi)$, where $\Phi(\phi) = e^{im\phi}/\sqrt{2\pi}$ (the $\hat{L}_z$ eigenvalue condition is just a simple ODE in ϕ), and $\Theta(\theta)$ is given by the associated Legendre polynomials $P_l^m(\cos\theta)$. The first few are:

$$Y_0^0 = \frac{1}{\sqrt{4\pi}}, \qquad Y_1^0 = \sqrt{\frac{3}{4\pi}}\cos\theta, \qquad Y_1^{\pm 1} = \mp\sqrt{\frac{3}{8\pi}}\sin\theta\, e^{\pm i\phi}.$$

- The spherical harmonics are orthonormal on the unit sphere:

$$\int_0^{2\pi}\int_0^{\pi} \left(Y_{l'}^{m'}\right)^* Y_l^m \,\sin\theta\, d\theta\, d\phi = \delta_{ll'}\,\delta_{mm'},$$

  and they form a complete basis for any square-integrable function of (θ, ϕ). This means any angular dependence of a wavefunction can be expanded in spherical harmonics; they are the spherical analog of the Fourier basis.

8.4 The Hydrogen Atom

- Now, we have gotten to our main task. To use quantum mechanics to solve the wave function for the hydrogen atom.

- The hydrogen atom is a single electron (mass m_e, charge $-e$) bound to a fixed proton (mass m_p, charge $+e$) by the Coulomb potential. Our Hamiltonian is $\hat{H} = \hat{p}_p^2/2m_e + p_e^2/2m_p - e^2/(4\pi\epsilon_0 r)$. Working in the center-of-mass frame, we get $\hat{H}_{rel} = \hat{p}^2/(2\mu) - e^2/(4\pi\epsilon_0 r) = -\hbar^2\nabla^2/2\mu - e^2/(4\pi\epsilon_0 r)$

- In spherical coords, we write $\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) - \frac{\hat{L}^2}{\hbar^2 r^2}$. Therefore, our Hamiltonian equation is

$$\hat{H} = -\frac{\hbar^2}{2m_e}\frac{1}{r^2}\frac{\partial}{\partial r}(r^2\frac{\partial}{\partial r}) + \frac{\hat{L}^2}{2m_e r^2} - \frac{e^2}{4\pi\epsilon_0 r}$$

- Our Hamiltonian commutes with both angular momentum operators because the Coulomb Potential is spherically symmetric (pretty obvious by the formula). Therefore, the energy eigenstates can be chosen to simultaneously diagonalize $\hat{H}$, $\hat{L}^2, \hat{L_z}$. Therefore, our PDE becomes separable: $\psi_{nlm}(r,\theta,\phi) = R_{nl}(r)Y_l^m(\theta,\phi)$. Subbing into the eigenvalue equation and using $\hat{L}^2 Y_l^m = \hbar^2 l(l+1)Y_l^m$, the angular parts cancel and we are left with our radial ODE for $R_{nl}(r)$.

8.4.1 The Radial Equation

- Subbing $\psi = R_{nl}(r)Y_l^m$ and simplifying with u(r)=rR(r), we get the radial equation:

$$-\frac{\hbar^2}{2m_e}\frac{d^2u}{dr^2} + \left[-\frac{e^2}{4\pi\epsilon_0 r} + \frac{\hbar^2}{2m_e}\frac{l(l+1)}{r^2}\right]u = Eu$$

- While this equation is very long, with the correct perspective, it becomes easier. This looks like a 1D Schrodinger equation in the variable r, with an effective potential: $V_{eff}(r) = -\frac{e^2}{4\pi\epsilon_0 r} + \frac{\hbar^2 l(l+1)}{2m_e r^2}$. The first term is the attractive Coulomb potential. The second is the centrifugal barrier.

- For bound states ($E < 0$), requiring $u(r) \to 0$, as $r \to 0$ and $r \to \infty$ (normalizability) quantizes energy. The analysis proceeds by examining asymptotic behavior at zero and infinity, then solving the remaining equation by power series. The series must truncate to remain normalizable, and this truncation condition introduces the principal quantum number $n = 1, 2, 3, ...$, with the constraint that $l = 0, 1, 2, ..., n-1$.

- So for a given n, there are n possible values of l, and for each l there are $2l+1$ values of m. The total degeneracy of the n-th energy level is: $\sum_{l=0}^{n-1}(2l+1) = n^2$

8.4.2 The Energy Spectrum and the Radial Wavefunctions

- Our quantize energies are:

$$E_n = -\frac{m_e}{2\hbar^2}\left(\frac{e^2}{4\pi\epsilon_0}\right)^2\frac{1}{n^2} \equiv -\frac{13.6,\text{eV}}{n^2}, \qquad n = 1, 2, 3, \ldots$$

- This is the Bohr formula, and there are several things to notice: The energies depend only on n, reflecting a deep SO(4) symmetry. The ground state energy $E_1 = -13.6eV$ sets the scale of atomic physics, and to ionize hydrogen we must supply the system with $13.6eV$. Moreover, the spacing between levels decreases as n increases: $\Delta E \to 0$ as $n \to \infty$, and above E=0 the spectrum becomes continuous (coulombic scattering states).

- The radial solution wavefunctions are in terms of the associated Laguerre polynomials: $R_{nl} = L_{n-l-1}^{2l+1}(2r/na_0)$, where a_0 is the Bohr radius (the natural length scale of hydrogen).

- Therefore, we have that the full, normalized eigenstates are: $\psi_{nlm}(r, \theta, \phi) = R_{nl}(r)Y_l^m(\theta, \phi)$. These are labeled by three quantum numbers, each with a physical meaning: n (principal; determines the energy of the state: $n \in \mathbb{N}$), l (azimuthal; determines total angular momentum: $l = 0, 1, ..., n-1$). m (magnetic; determines the z-component of angular momentum: $m = -l, ..., l$). This triplet (nlm) specifies which electron orbital an electron will be in.

8.5 The Spectrum and Atomic Transitions

- Electrons in hydrogen can transition between energy levels by absorbing or emitting a photon. Conservation of energy requires the photon energy to exactly match the level spacing:

$$E_\gamma = \hbar\omega = |E_{n'} - E_n| = 13.6eV\left(\frac{1}{n^2} - \frac{1}{n'^2}\right)$$

- This represents a transition from level n to level $n' > n$ (absorption) or $n' < n$ (emission). These are the spectral lines of hydrogen, organized into series named by their final level: Lyman (n'=1: to ground state, emission of UV photons), Balmer (n'=2: to 1st excited state, emission of visible light), and Paschen (n'=3: to 2nd excited state, infrared)

9 Commutation Relations & Time Evolution

- In the previous section, we established the power of operator algebra and the commutation relation as tools within quantum mechanics. Let's talk about them some more.
- The commutator is the central object in quantum mechanics that encodes how two observables relate to one another: whether they can be known simultaneously, whether one generates transformations that leave the other unchanged, and whether physical quantities are conserved over time.
- This section develops the general theory of commutators, makes precise the notion of compatible observables and complete sets thereof, and then applies that machinery to derive the Ehrenfest theorem and the full structure of time evolution for both operators and expectation values.

9.1 The Commutator

- The commutator of two operators $\hat{A}, \hat{B}$ is defined as: $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$. If the commutator is zero, we say the operators commute; if non-zero, they do not. The commutator measures the degree to which the order of applying two operators "matters".
- We have several algebraic identities as a direct result of the definition: Antisymmetry, Multilinearity, Product Rule (like we use in calculus), and the Jacobi identity.
- We already established a few key commutation relations:

$$[\hat{x}_i, \hat{p}j] = i\hbar\delta ij, \qquad [\hat{x}_i, \hat{x}_j] = 0, \qquad [\hat{p}_i, \hat{p}_j] = 0$$

$$[\hat{L}_i, \hat{L}j] = i\hbar, \varepsilon ijk\hat{L}_k, \qquad [\hat{L}^2, \hat{L}_i] = 0$$

$$[\hat{a}, \hat{a}^\dagger] = 1, \qquad [\hat{H}, \hat{a}] = -\hbar\omega\hat{a}, \qquad [\hat{H}, \hat{a}^\dagger] = +\hbar\omega\hat{a}^\dagger$$

- All of the physics of spin precession, angular momentum, and the harmonic oscillator spectrum emerged from examining these relations.

9.2 Commutators and the Uncertainty Principle

- We established in Postulate V that for any two observables $\hat{A}, \hat{B}$, they must satisfy the generalized uncertainty principle: $\Delta A \Delta B \geq \frac{1}{2}|\langle \hat{A}, \hat{B} \rangle|$. This establishes a universal limit on the structure of the global Hilbert space, not a comment on experimental uncertainty.
- We noted earlier, the most well-known of these uncertainties is the Heisenberg uncertainty principle: $\Delta x \Delta p \geq \frac{\hbar}{2}$.
- More generally, any pair of non-commuting observables has a fundamental lower bound on the product of their uncertainties, set by the expectation of the commutators in a given state.

9.3 Compatible Observables & Observable Sets

- In the previous section, we discussed non-commuting observables and how uncertainty relations emerge. Lets talk now about what happens when two operators do commute.
- Two Hermitian operators commute iff they share an eigenbasis. We touched on this statement in the previous set of notes, but as a quick run through of the proof: If they share a common eigenbasis $\{|n\rangle\}$ with $\hat{A}|n\rangle = a_n|n\rangle$ and $\hat{B}|n\rangle = b_n|n\rangle$, then for any state,$|\psi\rangle$, we have $[\hat{A}, \hat{B}]|\psi\rangle = \sum c_n(a_n b_n - b_n a_n)|n\rangle = 0$. For the converse statement: if $[\hat{A}, \hat{B}] = 0$ and $|a_n\rangle$ is an eigenstate of $\hat{A}$ (by spectral theorem), then $\hat{B}\hat{A}|a_n\rangle = a_n\hat{B}|a_n\rangle = \hat{A}(\hat{B}|a_n\rangle)$, so $\hat{B}|a_n\rangle$ is also an eigenstate of $\hat{A}$ with the same eigenvalue. This means that within each eigenspace of $\hat{A}$, we can diagonalize $\hat{B}$, and repeating this argument can build a simultaneous eigenbasis.
- From the uncertainty principle, compatible observables can be measured simultaneously with arbitrary precision. Measuring $\hat{A}$ first, then $\hat{B}$ (or vice versa) leaves the state unchanged, since we collapse down to a simultaneous eigenstate.

- Incompatible operators can never be diagonalized simultaneously. Measuring $\hat{A}$ first collapses the wavefunction to be in an eigenstate of $\hat{A}$, which will usually be a superposition of eigenstates of $\hat{B}$. This is why in the Stern-Gerlach experiment, measuring X to be $|+\rangle_x$, then measuring the state in the Z basis, may yield you (if you measure X again) $|-\rangle_x$. The act of measurement on one disturbs the other.

9.3.1 Complete Sets of Commuting Observables

- In many systems, a single observable is not enough to uniquely label every state in the Hilbert space. For example, in our hydrogen atom example, we had (in the end) the labels n, l, m corresponding to the set of operators $\{\hat{H}, \hat{L}^2, \hat{L}_z\}$. Here, each observable in this set commutes with the others, and if we were to remove any operator, we would lose information about our system. This is an example of a **Complete Set of Commuting Observables**

- A CSCO is a set of mutually commuting Hermitian operators $\{\hat{A}, \hat{B}, \hat{C}, ...\}$ such that specifying a simultaneous eigenvalue for each operator in the set (we call these labels) uniquely determines an eigenstate within our Hilbert space (since specifying a greater and greater number of labels projects into deeper and deeper subspaces of the Hilbert Space).

- Looking back to the hydrogen atom, the CSCO of $\{\hat{H}, \hat{L}^2, \hat{L_z}\}$ uniquely labels each state ψ_{nlm}. Adding spin would extend this to $\{\hat{H}, \hat{L}^2, \hat{L_z}, \hat{S}_z\}$ to resolve spin degeneracy within the atom (we will address this later).

- Something to notice: often, the choice of CSCO is not unique. For the hydrogen atom, we could equally use $\{\hat{H}, \hat{L}^2, \hat{L_x}\}$ instead. Both uniquely label states, just with different bases (we choose z based on precedent).

9.4 Constants of Motion

- An observable is called a **Constant of Motion** if its expectation value is invariant under time for every state in the Hilbert space. From postulate III, the state evolves as $i\hbar\frac{d|\psi\rangle}{dt} = \hat{H}|\psi\rangle$. So we get the derivative of the expectation value of $\hat{A}$ is:

$$\frac{d}{dt}\langle\hat{A}\rangle = \frac{d}{dt}(\langle\psi|\hat{A}|\psi\rangle) = \left\langle\frac{\partial\hat{A}}{\partial t}\right\rangle + \frac{i}{\hbar}\langle[\hat{A},\hat{H}]\rangle$$

- This is the **Ehrenfest theorem** of quantum mechanics. The first term accounts for explicit time dependence in the operator, and the second term accounts for the dynamical evolution driven by the Hamiltonian.

- For time-dependent operators, we simply get $\frac{d}{dt}\langle\hat{A}\rangle = \frac{i}{\hbar}\langle[\hat{A},\hat{H}]\rangle$. This directly shows that if an operator commutes with the Hamiltonian, then its eigenstates are constants of motion (they will only oscillate with phase, but will never change probability).

- We have already seen an example in the spin precession example. We oriented a constant magnetic field in the z-direction. In the z-spin basis, an eigenstate's probability (in an arbitrary superposition) was time independent (oscillated in phase) since $[\hat{S}_z, \hat{H}] = 0$, but in the x-spin basis, each eigenstate's probability (in an arbitrary superposition) was time dependent since $[\hat{S}_x, \hat{H}]$.

- Often, we refer to the eigenstates related to constants of motion as **Stationary States**.

9.5 The Heisenberg Picture

- There are two main perspectives in the quantum world: the Heisenberg Picture and the Schrodinger Picture, named after their two heavyweight champions. So far, we have been working with the Schrodinger picture (wavefunctions and quantum states evolve in time; operators are constant).

- The Heisenberg picture is an equivalent formulation in which the state vector is fixed, and operators carry the time dependence. The two are related by a unitary transformation: $\hat{A}_H(t) = \hat{U}^\dagger(t)\hat{A}_S\hat{U}(t)$, where $\hat{U}(t) = e^{-i\hat{H}t/\hbar}$ is the time evolution operator (from the previous notes) and $\hat{A}_S$ is the Schrodinger picture operator. We can quite easily see that expectation values are consistent in both pictures.

- In the Heisenberg picture, the equation of motion is the full Ehrenfest theorem, but in the Schrodinger picture, we drop the time-dependence of the operator term (we assume constant operators in this perspective). These are the quantum analog of Hamilton's equations of motion in classical mechanics (we recover classical dynamics in the $\hbar \to 0$ limit).

- Additionally, we should notice that in the Heisenberg picture, the canonical relations are preserved at all times by virtue of the time evolution operator being unitary. We will explore more about unitary operators in Section 11.

- Applying the Ehrenfest theorem to position and momentum for a particle in a potential, $V(\hat{x})$, we get:
$$\frac{d\hat{x}_H}{dt} = \frac{i}{\hbar}[\hat{H}, \hat{x}_H] = \frac{\hat{p_H}}{m}, \ \frac{d\hat{p}_H}{dt} = \frac{i}{\hbar}[\hat{H}, \hat{p_H}] = -\frac{\partial V}{\partial \hat{x}_H}$$

- These are operator equations, but taking the expectation values immediately recovers Newton's second law! Quantum mechanics reproduces classical mechanics at the level of expectation values. For another example, if we were to do a similar situation for the quantum harmonic oscillator, we see that the expectation value of position oscillates sinusoidally.

10 Spin and Angular Momentum Addition

- From here, there are two directions to proceed: Angular Momentum Addition or Transformations and Symmetries. These explore two aspects of the section prior: Commuting Sets of Observables and Time Evolution (respectively). We will proceed with the first option to finish our angular momentum saga.

- Up to this point, we have treated orbital angular momentum $\hat{L}^2$ and spin angular momentum $\hat{S}$ as two separate objects: acting on spatial wave functions via spherical harmonics and finite-dimensional spin spaces, respectively.

- In reality, a particle carries both, and many physically important phenomena rely on us to combine them into a single total angular momentum. We will develop this theory in this section.

10.1 The Total Angular Momentum

- Let us start with the most generic consideration of angular momentum addition. Consider a system with two independent angular momenta $\hat{J}_1, \hat{J}_2$. The operators act on different spaces, so they commute with each other, but abide by their own angular momentum commutation relation: $[\hat{J}_1^{(i)}, \hat{J}_2^{(j)}] = 0, \ \forall \ i, j$, but $[\hat{J}_a^{(i)}, \hat{J}_a^{(j)}] = i\hbar\epsilon_{ijk}\hat{J}_k$

- Total angular momentum is defined as $\hat{J} = \hat{J}_1 + \hat{J}_2$, or more precisely $\hat{J} = \hat{J}_1 \otimes \hat{I}_2 + \hat{I}_1 \otimes \hat{J}_2$, since they act on different parts of our total Hilbert space. (Postulate IV). Because each $\hat{J}_i$ satisfies the angular momentum algebra, and because the two distinct angular momenta commute with each other, the total $\hat{J}$ satisfies the same algebra. So $\hat{J}$ acts like its own angular momentum operator with all the properties derived in Section 7, with eigenvalues $\hbar^2 j(j+1)$ and $\hbar m_j$ respectively.

- Something to notice: we now have two natural bases for the combined system represented by two equivalent sets of commuting observables:
  - The uncoupled basis $\{|j_1, m_1\rangle \otimes |j_2, m_2\rangle = |j_1, m_1, j_2, m_2\rangle\}$ in which our set of commuting observables are $\{\hat{J}_1^2, \hat{J}_2^2, \hat{J}_{1z}, \hat{J}_{2z}\}$
  - The coupled basis $\{|j_1, j_2, j, m_j\rangle\}$ in which our set of commuting observables are $\{\hat{J}_1^2, \hat{J}_2^2, \hat{J}^2, \hat{J}_z\}$

- Both are valid orthonormal bases for the same Hilbert space: the uncoupled basis thrives in problems where the operators don't interact, and the coupled (as the name suggests) thrive when the operators are coupled. The addition of angular momenta is the problem of finding the unitary change of basis between them.

10.2 The Allowed Values of Total Angular Momentum

- Given two systems with definite angular momentum quantum numbers j_1 and j_2, what are the possible values of the total quantum number j? The z-components add simply: $m_j = m_1 + m_2$. The maximum possible value of m_j is $m_1^{max} + m_2^{max} = j_1 + j_2$. Thus, the largest total angular momentum is $j_{max} = j_1 + j_2$.
- The minimum value follows from the requirement that the total number of states is preserved. The uncoupled basis has $(2j_1 + 1)(2j_2 + 1)$ states. For our coupled basis to have the same dimension, it must have the same count, with each j contributing 2j+1 states. We can show mathematically that this works iff j runs in integer steps from $|j_1 - j_2|$ to $j_1 + j_2$. This is the triangle rule: the total number of states in the coupled basis is:

$$\sum_{j=|j_1-j_2|}^{j_1+j_2} (2j+1) = (2j_1+1)(2j_2+1)$$

- Something to notice is that, in the uncoupled basis, we multiply the dimensionality of the two individual spaces (by virtue of the tensor product), and in the coupled basis, we add up a series of (2j+1) by virtue of the total Hilbert space being the direct sum of a series of smaller Hilbert spaces (representing each j degeneracy).

10.3 The Clebsh-Gordan Coefficients

- The change of basis between the uncoupled and coupled representations is given by the **Clebsch-Gordon coefficients**:

$$|j_1, j_2; j, m_j\rangle = \sum_{m_1,m_2} \langle j_1, m_1; j_2, m_2 | j_1, j_2; j, m_j\rangle \, |j_1, m_1; j_2, m_2\rangle$$

- The coefficient $\langle j_1, m_1; j_2, m_2 | j, m_j\rangle$ (dropping the j_1, j_2 labels for brevity) is CG coefficent. It is only non-zero when $m_1 + m_2 = m_j$ (by the requirement of $\hat{J}_z = \hat{J}_{1z} + \hat{J}_{2z}$), and the coefficients are real by phase convention
- The CG coefficients are computed by starting from the highest-weight state and applying the lowering operator $\hat{J}_- = \hat{J}_{1-} + \hat{J}_{2-}$ repeatedly, using the known action (from prior sections): $\hat{J}_- \, |j, m_j\rangle = \hbar\sqrt{j(j+1) - m_j(m_j-1)} \, |j, m_j - 1\rangle$
- The highest-weight state $|j, j\rangle = |j_1, j_1; j_2, j_2\rangle$ (all angular momenta maximally aligned) is unique and known. Lowering it generates all states in the $j = j_1 + j_2$ multiplet. The states for smaller j values are then constructed by requiring orthonormality (using the Gram-Schmidt process) with respect to the multiplets already found. There is also the obvious completeness requirement.
- For small values of j_1 and j_2, CG coefficients are tabulated. For large values, they are computed numerically when needed.

10.4 Addition of Two Spin-1/2 Particles

- One of the more important and common examples we can do is adding two spin-1/2 systems $j_1 = j_2 = 1/2$. This describes two electrons, a proton and an electron, or any two-qubit system with uncoupled basis $|\uparrow\uparrow\rangle$, $|\uparrow\downarrow\rangle$, $|\downarrow\uparrow\rangle$, $|\downarrow\downarrow\rangle$, where $|\uparrow\rangle = |+\rangle, |\downarrow\rangle = |-\rangle$.
- By the triangle rule, the allowed total spin values are $j = 0$ and $j = 1$, given 1+3=4 states total. The coupled states are composed of a triplet (j=1, symmetric) and a singlet (j=0, antisymmetric):

$$|1,1\rangle = |\uparrow\uparrow\rangle, \quad |1,0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle), \quad |1,-1\rangle = |\downarrow\downarrow\rangle, \quad |0,0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$$

- These were derived in the same procedure mentioned above. Something to notice: the triplet states are symmetric under exchange of the two particles (swapping particle 1 and particle 2 leaves the state unchanged), while the singlet is antisymmetric (it picks up a minus sign). This will become essential in Section 11 (Multi-Particle Systems) when we enforce the Pauli exclusion principle. For two electrons, the total state (spin $\times$ spatial) must be antisymmetric, so the spin part being symmetric (triplet) forces the spatial part to be antisymmetric, and vice versa.

10.5 Spin-Orbit Coupling

- Another key example of angular momentum addition is **Spin-Orbit Coupling**, a phenomenon that arises because in the electron's rest frame, the orbiting nucleus creates a magnetic field that interacts with the electron's magnetic moment.
- The spin-orbit Hamiltonian takes the form: $\hat{H}_{SO} = \frac{1}{2m_e^2c^2}\frac{1}{r}\frac{dV}{dr}\hat{L}\cdot\hat{S}$, where $V(r) = -e^2/(4\pi\epsilon_0 r)$ is the Coulomb potential. This is a perturbation on top of the hydrogen Hamiltonian.
- To evaluate, we use $\hat{J}^2 = (\hat{L}+\hat{S})^2 = \hat{L}^2 + 2\hat{L}\cdot\hat{S} + \hat{S}^2 \implies \hat{L}\cdot\hat{S} = \frac{1}{2}(\hat{J}^2 - \hat{L}^2 - \hat{S}^2)$. In the coupled basis $\{n, l, j, m_j\}$, which simultaneously diagonalizes $\{\hat{J}^2, \hat{L}^2, \hat{S}^2\}$ with eigenvalues:

$$\langle \hat{L}\cdot\hat{S}\rangle = \frac{\hbar^2}{2}[j(j+1) - l(l+1) - s(s+1)]$$

- For an electron $s = 1/2$, so $s(s+1) = 3/4$. For a given l, the total spin can be $j = l \pm 1/2$, giving two values:

$$\langle \hat{\mathbf{L}}\cdot\hat{\mathbf{S}}\rangle = \begin{cases} +\frac{\hbar^2}{2}, l & j = l + \frac{1}{2} \\ -\frac{\hbar^2}{2}(l+1) & j = l - \frac{1}{2} \end{cases}$$

- The spin-orbit correction therefore splits each hydrogen level (n, l) with l ¿ 0 into two sub-levels with $j = l + 1/2$ and $j = l - 1/2$. This is the origin of the fine structure of hydrogen, the small splitting of spectral lines observed when a hydrogen spectrum is resolved at high precision.

11 Transformations and Symmetries

- Now we begin with the other path left for us at the end of Section 9: exploring how transformations act on quantum states, and how symmetries constrain the structure of quantum mechanics.
- The unifying idea of this section is that every continuous symmetry of a physical system is associated with a conserved observable (**Noether's theorem**), and that observable is the generator of the corresponding transformations. In quantum mechanics, this connection is precise and algebraic, highlighting the fundamental requirement of symmetry in physics.

11.1 Unitary Operators as Transformations & Their Generators

- We already used this logic in the set of notes before, deriving the Schrodinger from physical principles, unitary operators, and inferences.
- In quantum mechanics, a physical transformation (rotations, translations, time shifts...) must map valid states to valid states. Specifically, it must preserve the norm of states (the sum of probabilities is 1) and preserve the inner product structure.
- In our notes on linear algebra, we commented that any map on a Hilbert space that preserves inner products is unitary ($\hat{U}\hat{U}^\dagger = \hat{I}$ by Wigner's theorem.
- Something to note: unitary operators generalize orthogonal matrices from linear algebra over real fields to complex Hilbert spaces. They are the quantum analogue of "rigid motions"; i.e., they move vectors around without stretching them.

- **One-Parameter Families of Unitary Operators** $U(\lambda)$ with $U(0) = \hat{I}$ and $U(\lambda_1)U(\lambda_2) = U(\lambda_1 + \lambda_2)$ describes a continuous transformation parameterized by a real number λ. For small λ, we can use a first-order Taylor expansion to analyze the structure of U:

$$U(\lambda) \approx \hat{I} - \frac{i}{\hbar}\lambda\hat{G}$$

- We call $\hat{G}$, a Hermitian operator, the **generator** of the transformation. The factor of $-i/\hbar$ is a conventional choice, making $\hat{G}$ Hermitian (and revealing observable structures) whenever $U(\lambda)$ is unitary.

- For finite λ, exponentiating the infinitesimal form gives: $U(\lambda) = e^{-i\lambda\hat{G}/\hbar}$. This is the central formula of this section. Every continuous transformation is an exponential of its generator.

11.2 Corresponding Symmetry Pairs

- We already encountered these constructions of generator and transformation pairs before, in Section 5, when we derived the momentum operator from translational invariance. We revisit it now in the language of generators.

- Let's define the spatial translation operator $\hat{T}(a)$ by its action on the position basis: $\hat{T}(a)\,|x\rangle = |x + a\rangle$, or for wavefunctions: $\hat{T}(a)\psi(x) = \psi(x - a)$. From the generator formula we have $\hat{T}(a) = e^{-ia\hat{p}/\hbar}$, and the generator of spatial translations is the momentum operator $\hat{p}$. This is not a definition of the momentum operator; it is the derivation of the momentum operator from symmetry principles.

- Something to notice: the canonical commutation relation $[\hat{x}, \hat{p}] = i\hbar$ can be recovered entirely from requiring that $\hat{T}(a)$ is unitary and that $\hat{x}$ transforms as $\hat{T}(a)\hat{x}\hat{T}(a) = \hat{x} + a\hat{I}$. The algebra of the transformation encodes the commutator.

- The rotation operator $\hat{R}(\hat{n}, \phi)$ rotates a quantum state by angle ϕ about the axis $\hat{n}$. By the same generator logic: $\hat{R}(\hat{n}, \phi) = e^{-i\phi\hat{n}\cdot\hat{J}/\hbar}$, where $\hat{J} = (\hat{J}_x, \hat{J}_y, \hat{J}_z)$ is the total angular momentum operator. Angular momentum is therefore the generator of rotations (crazy jump in logic,, I know).

- This tells us something profound about where the commutation relations of angular momentum come from. In Section 7, we stated them as postulated properties; now we see they follow from the group structure of rotations in 3D space (specifically, rotations about different axes do not commute). The algebra of rotations forces the algebra of their generators

- Something to notice: spin is the angular momentum associated with a representation of the rotation group that has no classical counterpart (a half-integer representation). The rotation operator for a spin-1/2 particle rotates the spinor, not a spatial wavefunction:

$$\hat{R}_{\hat{n}}(\phi) = e^{-i\phi\hat{n}\cdot\vec{\sigma}/2}$$

- This reveals a famous result: a spin-1/2 state acquires a minus sign under a 2π rotation. It takes a 4π rotation to return to the original state. This emerges from the topological property of SU(2) being a double cover of SO(3).

11.3 Symmetries of the Hamiltonian and Conservation Laws

- We can now see that Postulate III is a special case of the generator formalism. The time evolution operator $\hat{U}(t, t_0) = e^{-i(t-t_0)\hat{H}/\hbar}$ is a one-parameter unitary family in time. Additionally, we derived in prior notes that the time-dependent Schrodinger equation is just the infinitesimal version of this statement, exactly as the momentum operator arose from the infinitesimal translation operator.

- Something that you may be curious about is that all three fundamental transformations (translations, rotations, and time shifts) are coincidentally generated by the most fundamental observables of quantum mechanics. These are the generators by spacetime symmetry encoded by the very nature of the universe, and quantum mechanics simply inherits them.
- Fun fact: there is another type of symmetry called boost invariance stemming from symmetries in shifting inertial reference frames, but we don't really care about it or its generator for Quantum Mechanics.
- At the end of Section 9, we discussed that if an observable commutes with the Hamiltonian, it is a conserved quantity in time. This is simply the quantum statement of Noether's theorem: every continuous symmetry of $\hat{H}$ implies a conserved observable. This lets us determine what are "good" and "bad" eigenstate labels; e.g. n, l, m from the hydrogen atom.

12 Multi-Particle Systems

- With the symmetry framework of Section 11 in hand, we are now capable of fully examining Postulates IV and VI to develop quantum systems with more than one particle. This is where quantum mechanics truly diverges from the classical world, not just in its probabilism, but in the types of non-locality and correlations that multi-particle systems and statistical quantum mechanics can exhibit.

12.1 Two-Particle Hilbert Spaces

- The notion of jumping from a single quantum particle system to a many-particle system is intimidating. For all we know, there could be some mystical quantum bull crap property that makes all of our understandings crumble. Well, not entirely because, luckily, we have Postulate VI.
- Anyways, let's take a baby step into multi-particle systems with the simplest possible case. A two-particle system, where each particle is distinguishable.
- Consider two (distinguishable) particles, each with its own Hilbert space $\mathcal{H}_1$ and $\mathcal{H}_2$. By Postulate IV, the total Hilbert space is their tensor product: $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$.
- A basis for $\mathcal{H}$ is formed by taking all pairs of basis vectors from the two spaces: if $\{|a_i\rangle\}$ spans $\mathcal{H}_1$ and $\{|b_j\rangle\}$ spans $\mathcal{H}$. The dimension of $\mathcal{H}$ is $dim(\mathcal{H}_\infty) \times dim(\mathcal{H}_2)$ since tensor products multiply dimension (pretty self explanatory as to why).
- The most general state in $\mathcal{H}$ is $|\Psi\rangle = \sum c_{ij} |a_i\rangle \otimes |b_j\rangle$, with normalization $\sum |c_{ij}|^2 = 1$.
- An operator $\hat{A}$ that acts only on particle 1 is extended to the full spaces as $\hat{A} \otimes \hat{I}_2$, and vice versa. Two observables acting on different particles will always commute since they act on independent subspaces.
- For continuous variables, the two-particle wavefunction is $\Psi(x_1, x_2) = \langle x_1, x_2 | \Psi \rangle$, and the probability density for simultaneously finding particle 1 at x_1 and particle 2 at x_2 is $|\Psi(x_1, x_2)|^2$.
- Not so bad, right? We don't care how their wave functions interact so long as they are distinguishable and the Hamiltonian doesn't have a coupling term (like Spin-Orbit Coupling).

12.2 Separable and Entangled States

- Now, let's talk about the first bit of quantum weirdness when we consider multiple quantum particles. The chance that they will entangle. We have already discussed the idea of a Bell state, but let us revisit this notion.
- A state $|\Psi\rangle \in \mathcal{H}_1 \otimes \mathcal{H}_2$ is called **separable** (or a product state) if it can be written as a tensor product of individual single-particle states: $|\Psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle$. In a product state, the two particles are statistically independent: the measurement outcome on particle 1 carries no information about particle 2.

- A state that cannot be written as any such product is called **entangled**. The canonical example we have already mentioned is the Bell state: $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|{+}{+}\rangle + |{-}{-}\rangle)$. If we measure particle 1 and find $|+\rangle$, we will know the state of particle 2, regardless of the spatial separation between the two particles. Nothing like this occurs in classical mechanics due to the causality rule. Something to be aware of: entanglement is a property of the state, not the particle. Entanglement is created by interactions and by measurement.

- The four Bell states form a maximally entangled orthonormal basis for the two-qubit Hilbert space ($\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^2$): $|\Phi^\pm\rangle = \frac{1}{\sqrt{2}}(|{+}{+}\rangle \pm |{-}{-}\rangle)$, $|\Psi^\pm\rangle = \frac{1}{\sqrt{2}}(|{+}{-}\rangle \pm |{-}{+}\rangle)$

- These are the maximally entangled states in the sense that measuring either particle gives a completely random outcome, yet the outcomes are perfectly (anti-)correlated. We will return to these in Section 13, discussing the idea that, in actuality, the universe is non-local.

12.3 The Exchange Operator

- We will now start breaking down the other bit of quantum "weirdness."

- When dealing with two particles of the same species (two electrons, two photons, etc.), there is no physical observable that distinguishes particle 1 from particle 2; they are fundamentally identical. This forces us to ask: what are the allowed states?

- Let us define the exchange operator, $\hat{P}_{12}$ (not to be confused with the projection operator), by its action: $\hat{P}_{12}|\psi_1\rangle \otimes |\psi_2\rangle = |\psi_2\rangle \otimes |\psi_1\rangle$, extended linearly to all of $\mathcal{H}$. Since applying it twice returns it to the original state, $\hat{P}_{12}^2 = \hat{I}$, and its eigenvalues must satisfy $\lambda^2 = 1$. giving $\lambda = \pm 1$.

- $\hat{P}_{12}$ is the quantum analogue of parity, but for the particle-label space. The $+1$ eigenstates are symmetric under exchange; the -1 eigenstates are anti-symmetric.

- For identical particles, Postulate VI mandates that all physically realizable states must be eigenstates of $\hat{P}_{12}$: Bosons (integer spin) and Fermions (half-integer spin) respectively.

- Something to notice: the requirement that the full state is an eigenstate of $\hat{P}_{12}$ is much stronger than just saying the particles are indistinguishable. It restricts the entire Hilbert space to a particular subspace: the symmetric subspace for bosons and the antisymmetric subspace for fermions. Not all vectors in $\mathcal{H} \otimes \mathcal{H}$ are physically realizable for identical particles.

12.4 Identical Fermions and the Slater Determinant

- For two fermions occupying a single particle state $|\phi_a\rangle$ and $|\phi_b\rangle$, with $a \neq b$, the correctly anti-symmeterized two-particle state is: $|\Psi_F\rangle = \frac{1}{\sqrt{2}}(|\phi_a\rangle \otimes |\phi_b\rangle - |\phi_b\rangle \otimes |\phi_a\rangle)$. (The wavefunction in position space looks identical; simply swap kets for wavefunctions). Immediately, if $a = b$ (both fermions in the same single-particle state), then $\Psi_F = 0$. (Pauli Exclusion Principle)

- For N fermions occupying states $\phi_1, \phi_2, ..., \phi_N$, the generalization of the antisymmetric state is the **Slater determinant**:

$$\Psi_F(x_1, ..., x_N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \phi_1(x_1) & \phi_1(x_2) & \dots & \phi_1(x_N) \\ \phi_2(x_1) & \phi_2(x_2) & \dots & \phi_2(x_N) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_N(x_1) & \phi_N(x_2) & \dots & \phi_N(x_N) \end{vmatrix}$$

- The Slater determinant automatically enforces antisymmetry: swapping any two particle labels corresponds to swapping two columns of the determinant, which flips its sign. If any two rows are identical (two particles in the same state), the determinant vanishes identically. Note: the $\frac{1}{\sqrt{N!}}$ prefactor is the normalization factor.

12.5 Identical Bosons

- Now let's look at the fermion's sibling, the boson. For two bosons occupying single-particle states $|\phi_a\rangle, |\phi_b\rangle$, the correctly symmetrized two particle state is: $|\Psi_B\rangle = \frac{1}{\sqrt{2}}(|\phi_a\rangle \otimes |\phi_b\rangle + |\phi_b\rangle \otimes |\phi_a\rangle)$. Let us now look at the same special case $a = b$; $|\Psi_B\rangle = |\phi_a\rangle \otimes |\phi_a\rangle$ (renormalized), so any number of bosons may occupy the same single-particle state.
- Rather than excluding occupation, the symmetry of bosons enhances the probability of additional particles joining an already-occupied state
- The macroscopic occupation of a single quantum state by a large number of bosons is a Bose-Einstein condensate, a phase of matter that emerges when bosons are cooled below a critical temperature. In the condensed phase, quantum effects manifest at macroscopic scales.

12.6 Spin and Spatial Degrees of Freedom for Two Electrons

- A complete description of an electron requires both spatial and spin degrees of freedom. The full single-particle Hilbert space is $\mathcal{H}_{space} \otimes \mathcal{H}_{spin}$, and the two-electron Hilbert space is the tensor product of two such spaces.
- Postulate VI requires the total two-electron state (spatial $\times$ spin) to be anti-symmetric under exchange. This does not mean each part must be separately antisymmetric; they can combine in two ways: anti-symmetric spatial & symmetric spin or symmetric spatial & anti-symmetric spin. Therefore, (from Section 10.4) triplet spin implies an antisymmetric spatial wavefunction, and a singlet spin implies a spatially symmetric wave function.
- The physical consequence is that the spatial correlations between two electrons are directly tied to their spin state. This is known as the **exchange interaction**, and it is the quantum origin of Hund's rules and ferromagnetism

12.7 The Density Matrix and Reduced States

- When we have a composite system in state $|\Psi\rangle \in \mathcal{H}_1 \otimes \mathcal{H}_2$ but we want to describe the state of only subsystem 1, we can no longer, in general, use a state vector since the reduced state of subsystem 1 may not be a pure state.
- The density matrix $\hat{\rho}$ is a generalization of the state vector that can describe both pure states and statistical mixtures:
  - For a pure state $|\psi\rangle$: $\hat{\rho} = |\psi\rangle \langle\psi|$
  - For a mixed state (a classical ensemble of states $|\psi_i\rangle$ with probabilities p_i): $\hat{\rho} = \sum p_i |\psi_i\rangle \langle\psi_i|$
  - The expectation value of any observable in the state $\hat{\rho}$ is $\langle\hat{A}\rangle = Tr(\hat{\rho}\hat{A})$
- The reduced density matrix of a subsystem 1 is obtained by tracing over subsystem 2:

$$\hat{\rho}_1 = Tr_2(|\Psi\rangle \langle\Psi|) = \sum (\hat{I}_1 \otimes \langle b_j|) |\Psi\rangle \langle\Psi| (\hat{I}_1 \otimes |b_j\rangle)$$

- here $\{|b_j\rangle\}$ is any orthonormal basis for $\mathcal{H}_2$. This operation is called the partial trace. Something to note: for a product state, the reduced density matrix is pure. For an entangled state, tracing out subsystem 2 yields a mixed matrix, one that cannot be written as any projection onto a state.
- The degree of mixedness of $\hat{\rho}_1$ is a measure of entanglement, quantified by the **von Neumann entropy**: $S = -Tr(\hat{\rho}_1(log(\hat{\rho}_1)))$. A pure reduced state has $S = 0$ and a maximally entangled state (such as a Bell state) has $S = log(2)$. This entropy is precisely what makes the density matrix powerful at modelling noise, so we will see this throughout condensed matter and quantum information.

13 EPR Paradox and Schrodinger's Cat

- Now that we have seen some of the quantum weirdness at play in quantum mechanics, we should talk about the repercussions it has on physics. At its core, quantum mechanics agrees extremely well and not at all with reality. It predicts incredible physics at small scales, but in many ways, it breaks all senses of causality. But can we trust it and/or our physical sensibilities? No equations are introduced here that we have not already seen; the entire weight of the section is interpretational.
- The two thought experiments that anchor this section approach from opposite directions and pose theoretical challenges to the field as a whole.
- EPR (Einstein, Podolsky, Rosen, 1935) challenges quantum mechanics from outside: they argue the formalism must be incomplete because it seems to permit instantaneous information across space (violating causality). Their proposed remedy is so-called local variables. The resolution of EPR came in 1964 with Bell's theorem, when an experiment was designed and decisively answered it.
- Arguably, the most influential quantum thought experiment is Schrodinger's Cat (Schrödinger, 1935). It challenges quantum mechanics from inside: it asks what the superposition principle means when applied not just to electrons, but to macroscopic objects. It forces us to confront where (or whether) wavefunction collapse actually occurs.

13.1 The EPR Argument

- Einstein, Podolsky, and Rosen (1935) proposed that, if, without disturbing a physical system in any way, we can predict (with certainty) the value of a physical quantity, then there exists an element of physical reality corresponding to that quantity. A complete theory must account for every such element of reality.
- Long story short, we don't have a way to establish a method by which entanglement acts on particles over arbitrarily long distances. Thus, since quantum mechanics is non-local (no way to predict how information transfers between two entangled states), it is either false or incomplete.
- They thought of taking two electrons in a Bell state (maximally correlated) and separating them an arbitrarily long distance. As soon as we measure one, we know the other before information, traveling at the speed of light, can reach the other particle and collapse its wavefunction. Thus, wavefunction collapse is globally instantaneous.
- EPR's proposed remedy was to introduce local hidden variables (λ) not described by the wavefunction, that carry the "true" predetermined values of all observables. Something so out of sight that we cannot see it.
- Classically, this grievance is valid, but it was based on the necessary assumption of classical mechanics: we operate under **local realism**.

13.2 Bell's Theorem

- In 1964, John Bell showed that no sort of local hidden-variable theory can reproduce all the predictions of quantum mechanics. To do this, he derived an inequality (Bell inequality) that every local realistic theory must satisfy. Then, he showed that quantum mechanics predicts a violation of it. Experimentally, quantum mechanics came out on top.
- The most commonly tested form of Bell inequality is the **Clauser-Horne-Shimony-Holt (CHSH) inequality**:

$$S = |E(a,b) + E(a,b') + E(a',b) - E(a',b')| \leq 2$$

  where E(a,b) is the expectation value of the product of Alice's and Bob's measurement outcomes (± 1) when they choose measurement settings a (or a') and b (or b').

- Any local hidden-variable theory predicts that the parameter S can never exceed 2. Quantum mechanics retorts that for an entangled state $S = 2\sqrt{2}$
- Setup: A source prepares pairs of particles in an entangled quantum state and distributes one particle from each pair to two spatially separated observers, conventionally called Alice and Bob. At each measurement station, the observer independently and randomly selects one of several possible measurement settings and performs a projective measurement on the received particle.
- This procedure is repeated for many identically prepared pairs to estimate the joint probability distributions and correlation functions associated with different combinations of measurement settings. The measured correlations are then used to evaluate the Bell inequalities, whose violation is incompatible with any local hidden-variable theory. Nature is not locally realistic.

13.3 What Bell's Theorem Does and Doesn't Say

- It rules out any local and realistic (in the EPR sense) theory of quantum mechanics
- It doesn't rule out non-local hidden variable theories (like parts of QFT)
- It does not rule out theories that abandon realism while maintaining weak locality (e.g., the Copenhagen interpretation or the Many World theory)
- It doesn't allow for FTL signaling since, in any designed experiment, the observers need to exchange classical information.
- A final note: Bell's theorem doesn't tell us to abandon which (or both) locality and realism. Different theories make different choices as to which to maintain.

13.4 Schrodinger's Cat

- In the same year that EPR was proposed (1935), Schrodinger proposed a thought experiment to better understand Postulate II.
- The setup is infamous, but I will repeat for completeness: A cat is placed in a sealed box, with a quantum device housing a particle in (50/50) superposition, and a vial of poison. If the superposition collapses to one state, the beaker shatters, but if it collapses to the other, the beaker does nothing.
- We may say presumptuously, by postulate I and II, that the cat is simultaneously dead and alive. But this was a rhetorical question. In actuality, Schrodinger wants to reveal a conceptual gap: quantum mechanics does not specify when or why the superposition collapses into a definite outcome, just that it does. At what scales can we consistently observe quantum effects, and if we are made up of a bunch of quantum particles, why are we deterministic?

13.5 The Problem with Measurement

- The dead-and-alive cat is an illustration of the currently unresolved measurement problem of quantum mechanics. It is a dilemma (but with three options) of the following:
  - The wavefunction is a complete description of a physical system
  - The wavefunction always evolves according to the linear Schrodinger equation
  - Measurements always have definite (eigenstate) outcomes
- All of these statements can't be correct, and every modern interpretation of quantum mechanics (ways of providing a consistent ontological account of quantum mechanics) resolves this trilemma by rejecting one of the three statements.
- **Decoherence** is the most important physically grounded concept for understanding why macroscopic superpositions are not observed in practice. When a quantum system interacts with a large environment (heat bath, air molecules, photons), the system and environment become entangled. Thus, things become classical in the decoherence limit. It is also the biggest hurdle for quantum computing.

- There are several modern interpretations of physics that either reject this notion or entirely accept it (some of which you may know): the Copenhagen interpretation, the Many-Worlds interpretation, de Broglie-Bohm Pilot-Wave theory, Objective Collapse theories, etc.
- This is a rare situation in physics where foundational questions remain genuinely open.

14 Perturbation Theory and Adiabatic Processes

- A big issue physics faces, past basic mechanics and electromagnetism, is that most physical laws, while powerful, only have explicit solutions to a select few problems without requiring a computer to solve. The hydrogen atom, the harmonic oscillator, and the infinite square well are the rare exceptions in quantum mechanics. In nearly every realistic situation, the Hamiltonian is too complicated to yield closed-form eigenstates and eigenvalues.
- Perturbation theory gives a bit of relief by making it such that we can find higher-order solutions to perturbations of our system's Hamiltonian. In this process, we identify a solvable aspect (we can determine eigenstates and eigenvalues) of a system's Hamiltonian, $\hat{H}_0$, then add a small perturbation/correction with the operator $\hat{H}'$.
- The theory splits naturally into the time-independent case (how do energy levels shift when the Hamiltonian is perturbed?) and the time-dependent case (what is the probability of a transition between levels when a perturbation is switched on?). The adiabatic theorem then addresses what happens when the Hamiltonian changes, but changes slowly, which leads to a geometric result known as Berry's phase.

14.1 Time-Independent Non-Degenerate Perturbation Theory

- Let us first start with the most basic case of perturbation theory: a time-independent perturbation with a non-degenerate eigenspectrum. Here we write the full Hamiltonian as: $\hat{H} = \hat{H}_0 + \lambda\hat{H}'$, where we identified that $\hat{H}_0$ is the unperturbed Hamiltonian with known eigenstates and eigenvalues: $\hat{H}_0 |n^{(0)}\rangle = E_n^{(0)} |n^{(0)}\rangle$. Additionally, we should note that λ is a small dimensionless parameter ($\lambda << 1$) that controls the strength of our perturbation $\hat{H}'$. At the end, we will take the limit as $\lambda \to 1$, and hopefully nothing blows up in our face.
- We expand the exact eigenstates and eigenvalues as a power series in λ

$$|n\rangle = |n^{(0)}\rangle + \lambda |n^{(1)}\rangle + \lambda^2 |n^{(2)}\rangle + ...; \quad E_n = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + ...$$

- Substituting into our eigenvalue equation $\hat{H} |n\rangle = E_n |n\rangle$ and collecting terms order by order in λ generates a recursive set of equations for each order determined by previous orders. This is the core strategy of perturbation theory: reduce an intractable eigenvalue problem to a sequence of tractable ones.
  - (λ^0) At zeroth order, we recover our unperturbed eigenstate equation: $\hat{H}_0 |n^{(0)}\rangle = E_n^{(0)} |n^{(0)}\rangle$
  - (λ^1) At first order, we recover the eigenstate equation: $\hat{H}_0 |n^{(1)}\rangle + \hat{H}' |n^{(0)}\rangle = E_n^{(0)} |n^{(1)}\rangle + E_n^{(1)} |n^{(0)}\rangle$
  - (λ^1) At second order, we get the eigenstate equation: $\hat{H}_0 |n^{(2)}\rangle + \hat{H}' |n^{(1)}\rangle = E_n^{(0)} |n^{(2)}\rangle + E_n^{(1)} |n^{(1)}\rangle + E_n^{(2)} |n^{(0)}\rangle$
  - Iterate as many times as desired.
- Assuming we have already solved the base case, let us solve the first-order correction, then solve the second-order correction.

14.1.1 First-Order Corrections (Non-Degenerate, Time-Independent PT)

- At first order, we have the equation: $\hat{H}_0 |n^{(1)}\rangle + \hat{H}' |n^{(0)}\rangle = E_n^{(0)} |n^{(1)}\rangle + E_n^{(1)} |n^{(0)}\rangle$. We can solve this by applying $\langle n^{(0)}|$ (and using $\langle n^{(0)}| \hat{H}_0 = E_n^{(0)} \langle n^{(0)}|$ since $\hat{H}_0$ is Hermitian) gives the first-order energy correction: $E_n^{(1)} = \langle n^{(0)}|\hat{H}'|n^{(0)}\rangle$.
- The first-order energy shift is simply the expectation value of the perturbation in the unperturbed state. This is the most-used result in all of perturbation theory.
- To find the first-order state correction $|n^{(1)}\rangle$, project the first order equation onto $\langle k^{(0)}|$ for $k \neq n$:

$$\langle k^{(0)}|n^{(1)}\rangle = \frac{\langle k^{(0)}|\hat{H}'|n^{(0)}\rangle}{E_n^{(0)} - E_k^{(0)}} \implies |n^{(1)}\rangle = \sum_{k\neq n} \frac{\langle k^{(0)}|\hat{H}'|n^{(0)}\rangle}{E_n^{(0)} - E_k^{(0)}} |k^{(0)}\rangle$$

- Note: this expression diverges if $E_n^{(0)} = E_k^{(0)}$ for some $k \neq n$. That is, if the unperturbed level is degenerate. This is not a physical divergence but a signal that we have chosen the wrong zeroth-order basis. Degenerate perturbation theory (we will come back to this in a moment).
- Now, we have all the pieces to solve the second-order correction.

14.1.2 Second-Order Corrections (Non-Degenerate, Time-Independent PT)

- Our second-order equation from above is $\hat{H}_0 |n^{(2)}\rangle + \hat{H}' |n^{(1)}\rangle = E_n^{(0)} |n^{(2)}\rangle + E_n^{(1)} |n^{(1)}\rangle + E_n^{(2)} |n^{(0)}\rangle$. Doing the same procedure with applying $\langle n^{(0)}|$ yields:

$$E_n^{(2)} = \sum_{k\neq n} \frac{|\langle k^{(0)}|\hat{H}'|n^{(0)}\rangle|^2}{E_n^{(0)} - E_k^{(0)}}$$

- Before we write our eigenstate equations, a few notes. States with $E_k^{(0)} < E_n^{(0)}$ contribute negatively (they lower the energy), and states with greater energy eigenvalues increase the energy. For the ground state, all terms in the sum have $E_k^{(0)} > E_0^{(0)}$, so the perturbation always lowers the ground state energy at second order. This is the quantum analog of the variational principle. The correction is suppressed when the energy gap is large and sharply spikes when the gap is small.
- The perturbation expansion is controlled by the ratio. If this ratio is not small, higher-order terms will cause the perturbation to break down and diverge.
- Doing the same as for the first order, we get a very nasty eigenstate equation, but we will never use it. I am writing it below it for the sake of completeness:

$$|n^{(2)}\rangle = \sum_{m\neq n} \left[\sum_{k\neq n} \frac{V_{mk}V_{kn}}{\left(E_n^{(0)} - E_m^{(0)}\right)\left(E_n^{(0)} - E_k^{(0)}\right)} - \frac{V_{nn}V_{mn}}{\left(E_n^{(0)} - E_m^{(0)}\right)^2} \right] |m^{(0)}\rangle - \frac{1}{2} \sum_{m\neq n} \frac{|V_{mn}|^2}{\left(E_n^{(0)} - E_m^{(0)}\right)^2} |n^{(0)}\rangle .$$

- Here, $V_{mn} = \langle m^{(0)}|\hat{H}'|n^{(0)}\rangle$.

14.2 Degenerate Time-Independent Perturbation Theory

- When several unperturbed states share the same energy, say $\left\{|n_1^{(0)}\rangle, |n_2^{(0)}\rangle, \ldots, |n_m^{(0)}\rangle\right\}$, $\hat{H}_0 |n_i^{(0)}\rangle = E_n^{(0)} |n_i^{(0)}\rangle$, the non-degenerate formulas break down because the denominators vanish whenever $|n_k^{(0)}\rangle$ lies within the degenerate subspace.
- The resolution is to find the correct zeroth-order states: linear combinations of the degenerate eigenstates that diagonalize $\hat{H}'$ within the degenerate subspace. These are the eigenstates of the degenerate-subspace matrix (sometimes called the W-matrix), $W_{ij} = \langle n_i^{(0)}|\hat{H}'|n_j^{(0)}\rangle$.

- The eigenvalues of W are the first-order energy corrections $E_n^{(1)}$, and the corresponding eigenvectors determine the correct zeroth-order states. Once this basis is chosen, the non-degenerate perturbation formulas apply for all higher-order corrections.
- Degenerate perturbation theory can be viewed as the problem of finding the complete set of commuting observables (CSCO) for the perturbed Hamiltonian. The perturbation $\hat{H}'$ generically breaks the degeneracy, lifting the degenerate levels apart, but only along directions in the degenerate subspace that $\hat{H}'$ distinguishes. This simply means that not every linear combination of degenerate states is affected differently by the perturbation.

14.3 Time-Dependent Perturbation Theory

- Now, suppose the perturbation is suppose the perturbation is switched on at $t = 0$ and varies in time: $\hat{H}(t) = \hat{H}_0 + \hat{H}'(t)$. The question changes; rather than asking how energy eigenstates and values shift, we ask what the probability of transition from an initial eigenstate of $\hat{H}_0$ to a final eigenstate is after time t.
- This entire section is a more convoluted way to say to do a form of Fourier analysis.
- We work in the interaction picture (Dirac picture), which factors out the fast oscillations of $\hat{H}_0$. Define the state in the interaction picture as: $|\psi_I(t)\rangle = e^{i\hat{H}_0 t/\hbar}\,|\psi(t)\rangle$.
- The equation of motion in this picture is: $i\hbar\frac{d}{dt}\,|\psi_I(t)\rangle = \hat{H}'_I(t)\,|\psi_I(t)\rangle$, where

  $\hat{H}'_I(t) = e^{i\hat{H}_0 t/\hbar}\hat{H}'(t)e^{-i\hat{H}_0 t/\hbar}$ is the perturbation in the interaction picture. The full complexity of $\hat{H}_0$ has been absorbed into the operators, leaving only $\hat{H}'$driving the evolution of the state coefficients.
- Expanding the state in the unperturbed basis,$|\psi_i(t)\rangle = \sum_n c_n(t)\,|n\rangle$, and expanding to first order in $\hat{H}'$ with initial condition $c_n(0) = \delta_{ni}$:

$$c_f^{(1)}(t) = \frac{i}{\hbar}\int_0^t \langle f|\hat{H}'(t')|i\rangle\, e^{i\omega_{fi}t'}dt',$$

- Here, $\omega_{fi} = (E_f - E_i)/\hbar$ is the Bohr frequency between states $|i\rangle$ and $|f\rangle$. The transition probability to the first order is:

$$P_{i\to f}(t) = |c_f^{(1)}(t)|^2 = \frac{1}{\hbar^2}|\int_0^t \langle f|\hat{H}'(t')|i\rangle\, e^{i\omega_{fi}t'}dt'|^2$$

14.4 Fermi's (Dirac's) Golden Rule

- It's treat others the way you want to be treated... Actually, it's about transitions under a constant perturbation, but the first one is still good advice.
- One of the most important applications of perturbation theory is the solution to a constant perturbation $\hat{H}'$ switched on at $t = 0$: $\hat{H}'(t) = \hat{H}'$ for $t > 0$. The integral gives: $c_f^{(1)}(t) = -\frac{i}{\hbar}\langle f|\hat{H}'|i\rangle\int_0^t e^{i\omega_{fi}t'}dt' = -\frac{i}{\hbar}\langle f|\hat{H}'|i\rangle\frac{e^{i\omega_{fi}t}-1}{i\omega_{fi}}$
- The transition probability: $P_{i\to f}(t) = \frac{|\langle f|\hat{H}'|i\rangle|^2}{\hbar^2}\cdot\frac{4\sin^2(\omega_{fi}t/2)}{\omega_{fi}^2}$
- As $t\to\infty$, the factor $4sin^2(\omega_{fi}t/2)/\omega_{fi}^2$, becomes sharply peaked around $\omega_{fi} = 0$ (i.e. $E_f = E_i$) and, in the sense of distribution, $4sin^2(\omega_{fi}t/2)/\omega_{fi}^2 \to 2\pi t\delta(\omega_{fi})$. Since $\omega_{fi} = \frac{E_f - E_i}{\hbar}$, we have $\delta(\omega_{fi}) = \hbar\delta(E_f - E_i)$.
- For transitions into a continuum of final state with density of state $\rho(E_f)$, integrating over final states gives the transition rate:

$$\Gamma_{i\to f} = \frac{dP_{i\to f}}{dt} = \frac{2\pi}{\hbar}|\,\langle f|\hat{H}'|i\rangle\,|^2\rho(E_f)$$

- This is dubbed as Fermi's Golden Rule. (Despite the name, the result was derived primarily by Dirac; Fermi popularized its use and coined the term "golden rule.") A notable feature is that the delta function $\delta(E_f - E_i)$ enforces energy conservation. For a time-independent perturbation, only transitions satisfying $E_f = E_i$ survive in the long-time limit, and transitions where these energies aren't equal are suppressed in the long run.

- For a sinusoidal perturbation, $\hat{H}'(t) = \hat{V}e^{-i\omega t} + \hat{V}^\dagger e^{i\omega t}$, the same calculation yields two resonance conditions

$$\frac{2\pi}{\hbar}\left|\langle f|\hat{V}|i\rangle\right|^2 \rho(E_f)\delta(E_f - E_i - \hbar\omega) \qquad \text{(absorption)},$$

and

$$\frac{2\pi}{\hbar}\left|\langle f|\hat{V}^\dagger|i\rangle\right|^2 \rho(E_f)\delta(E_f - E_i + \hbar\omega) \qquad \text{(stimulated emission)}.$$

- The system absorbs energy ($\hbar\omega$) when $E_f = E_i + \hbar\omega$, and emits energy ($\hbar\omega$) when $E_f = E_i - \hbar\omega$. These resonance conditions, together with the symmetry properties of the matrix elements ($\langle f|\hat{V}|i\rangle$), lead to the spectroscopic selection rules and provide the quantum foundation of things like laser physics.

- Fermi's Golden Rule gives the transition rate, but the matrix element $\langle f|\hat{H}'|i\rangle$ may be zero for symmetry reasons, even before any calculation. When it does, the transition is forbidden (in first order). The conditions under which matrix elements vanish are called **selection rules**.

14.5 The Adiabatic Theorem

- Now we will consider a different type of problem, one where the Hamiltonian changes in time, but very slowly (a concept not foreign to those who have worked with angle action coordinates in the classical sense). Formally, we say the Hamiltonian varies on a timescale that is much larger than the internal timescales of the system (e.g. a driven classical harmonic oscillator whose driving frequency is much smaller than the natural frequency of the spring).

- The adiabatic theorem states: if a system starts in the n-th eigenstate of the Hamiltonian at time $t = 0$ (and the Hamiltonian evolves at a sufficiently slow rate), then the system remains in the n-th eigenstate throughout the evolution, even as the state deforms (without making transitions to other levels).

- This is in stark contrast to the derivations of time-dependent perturbation theory, where we turn on a perturbation spontaneously. If the Hamiltonian changes suddenly, the state has no time to adjust and remains in its initial form, which is generally a superposition of the new eigenstates. The adiabatic theorem approaches this problem from the other limit of a perturbation to our system.

14.6 Berry's Phase

- The adiabatic theorem says that the system stays in the n-th eigenstate, but it doesn't say anything about the phase accumulated along the way. We might expect only the usual **dynamical phase**: $e^{-i\int E_n(t')dt'/\hbar}$. Remarkably, there is an additional geometric phase that depends only on the path traced in parameter space (an inherently topological quantity that has a non-abelian variant used as a foundation in Topological Quantum Computing).

- Suppose the Hamiltonian depends on a set of slowly varying parameters given by $R(t)$. After one cycle, $R(T) = R(0)$, the n-th eigenstate returns to itself, but acquires a phase:

$$|\psi(T)\rangle = e^{i\gamma_n} e^{-\frac{i}{\hbar}\int_0^T E_n(t)dt}\,|n(R(0))\rangle$$

- The first exponential is the dynamical phase (the usual time-average of the energy). The second is Berry's phase:

$$\gamma_n = i\oint_C \langle n(R)|\nabla_R|n(R)\rangle \cdot dR$$

- Here we are taking the integral around the closed loop C in parameter space traced by R(t), and ∇_R represents the covariant derivative (encoding the curvature) of this parameter space.
- Some notes: Our Berry phase is real despite the factor of i because $\langle n|\nabla|n\rangle$ is purely imaginary. More importantly, as we mentioned, the factor is topological and is directly proportional to the winding number of the loop, making the phase resistant to perturbations (highlighting the nature of the adiabatic theorem). If too quick a perturbation were to occur, this loop might break, thus losing this property.
- As touched upon in its introduction, this concept is more than a curiosity; it underlies concepts such as the **Aharonov-Bohm effect**, **the Quantum Hall Effect**, and quantum gates in topological quantum computing.

15 A Few Words on Relativistic Quantum Mechanics

- Everything thus far has been non-relativistic (luckily). We have the Schrodinger equation that treats time as a "better" parameter than space and assumes particle number is conserved, both of which are assumptions that break down at higher energies. A complete quantum theory must reconcile quantum mechanics with special relativity. This section serves as a brief hint at that larger framework.
- The first attempt to write a relativistic wave equation was the **Klein-Gordon equation**, obtained from the equation $E^2 = (pc)^2 + (mc^2)^2$ and promoting $E \to i\hbar\partial_t$ and $p \to -i\hbar\nabla$:

$$-\hbar^2 \frac{\partial^2 \psi}{\partial t^2} = (-\hbar^2 c^2 \nabla^2 + m^2 c^4)\psi$$

- This equation is Lorentz covariant, space and time enter symmetrically, as relativity demands. However, it is second-order in time (two partial derivatives with respect to time), which leads to a probability density that is not positive-definite. It cannot be consistently interpreted as a single-particle wavefunction and is better understood as a field equation for spin-0 particles (e.g. the Higgs boson).
- Dirac resolved this by insisting on an equation that is first order in both space and time (Veritasium made a great video on this, go check it out), which requires the form $\hat{H} = c\alpha \cdot \hat{p} + \beta mc^2$, with α, β are matrices that satisfy $\{\alpha^i, \alpha^j\} = 2\delta^{ij}$, $\{\alpha^i, \beta\} = 0$, and $\beta^2 = 1$. The smallest matrices satisfying these anticommutation relations (here we use $\{\}$ as an anti-commutator; instead of a minus in the commutator formula, there is a plus) are 4×4 matrices, which means the Dirac equation describes a 4-component wavefunction. Two correspond to the spin states of the electron, and the other two correspond to anti-particles, so the Dirac equation predicts the existence of antimatter.
- The Dirac equation has the form(s) (the latter is the expanded covariant formulation):

$$(i\partial\!\!\!/ - m)\psi(x) = (i\hbar\gamma^\mu \nabla_\mu - mc)\psi = 0$$

- The Dirac equation also predicts the correct magnetic moment of the electron, giving $g = 2$ exactly at tree level (the small deviation from this, $g - 2 \approx a/\pi$, (a is the Pauli anomaly) is a quantum field theory correction).
- Moreover, we can recover the Schrodinger equation at leading powers of v/c
- Something to note: even the Dirac equation is ultimately an approximation. At high enough energies, pair creation and annihilation become possible. This is why the Dirac equation is only a stepping stone to the massive framework that is Quantum Field Theory. Quantum electrodynamics (QED), the QFT of electrons and photons, is the most precisely tested physical theory ever constructed, and the full Standard Model of particle physics, the most successful equation in physics, is a relativistic quantum field theory based on the foundations we have laid out in this paper.

16 Conclusion

This has been a massive paper and a large undertaking. I hope you have enjoyed the material and the fashion in which it was presented. If you notice any errors, feel free to contact me (I am human and can make mistakes). The material covered here should be able to help you excel in any undergraduate Quantum Mechanics class, and if there is anything I should add, feel free to contact me. Thank you for reading!